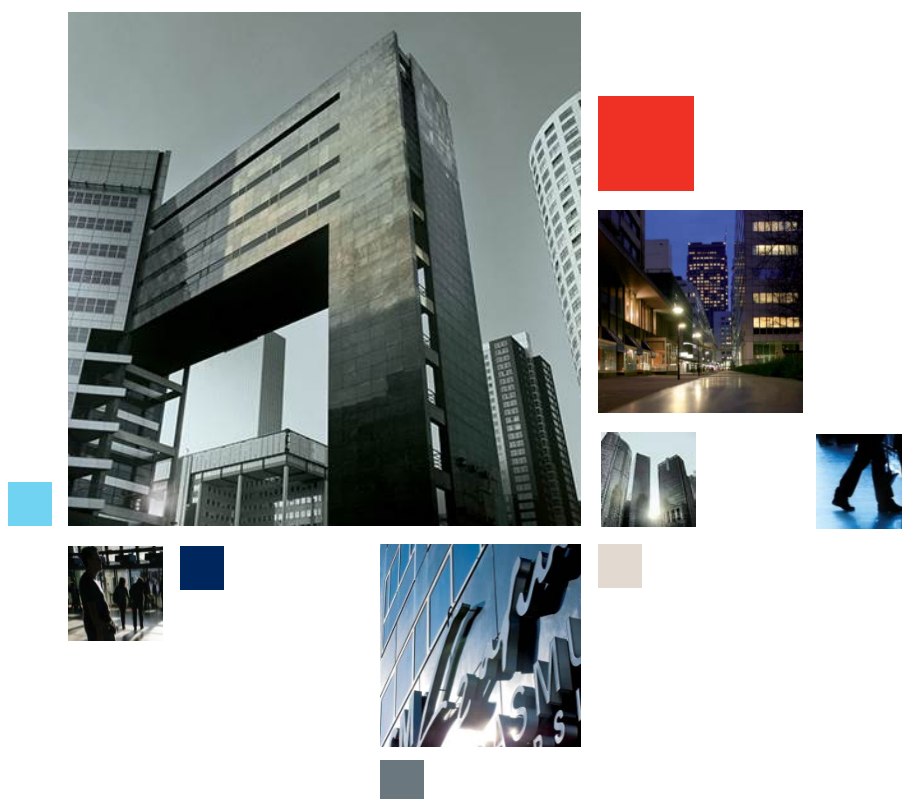


ROTTERDAM SCHOOL OF MANAGEMENT ERASMUS UNIVERSITY

A STUDY OF DIFFERENCES IN RETURNS BETWEEN LARGE AND SMALL COMPANIES IN EUROPE

Research Commissioned by Duff & Phelps LLC.



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1.1 Introduction

The size effect included in average realized equity returns is a widely examined phenomenon. Numerous studies analyzing US equity returns find that throughout the past decades, small capitalization ("small cap") shares have earned greater returns, on average, than large capitalization ("large cap") shares (see van Dijk (2011) for a review of these studies), suggesting that small firms have a higher cost of equity. In fact, these studies show that, depending on sample selection procedures, research period, and (sorting) methodology, the estimated monthly return difference between the largest cap and smallest cap shares may range from approximately 0.4 percent to almost 2.5 percent. Possible rational explanations for these observed size effects are that (1) liquidity risk varies with firm size, (2) firm size is a proxy for other priced (but unobservable) risk factors, or (3) investor preferences or recognition depend on firm size. Not surprisingly, a number of researchers have suggested that the observed size effect is an artifact of the data, partly motivated by the fact that most US studies examine overlapping datasets.

To test the robustness of US findings as well as to assist analysts in estimating the cost of equity in non-US markets, some researchers have attempted to investigate the size effect in samples of non-US shares. Many of these studies, especially those focusing on a single country, suffer from small sample size problems. In addition, studies examining groups of countries during the 1970s and 1980s may have been adversely affected by the historic lack of integration among some or all of the share exchanges in the sample. It is not inconceivable for the risk differential between small and large company shares to differ in an internationally diversified portfolio relative to a segregated portfolio of local company shares. This may occur, for example, when a lack of diversification opportunities in segregated markets makes investors averse to the higher idiosyncratic risk of small cap shares. Further, size effects observed in cross-

country studies could in fact be country effects if firm size systematically varies across countries (van Dijk, 2011). In this study, we re-assess the existence of the size effect outside the US using a large sample of Western European shares during the period 1990 - 2013, a time in which the European economies and share exchanges were largely and increasingly integrated (see e.g., Freimann, 1998). Specifically, our study is organized in two parts. In the first part of our study, we examine the size effect in the pooled sample, treating all European exchanges as a single integrated market. In the second part, we split up our sample into potentially more homogeneous geographical regions. This analysis helps not only to illustrate the consistency of the size effect across countries, but also to explore the influence of country factors on our pooled analysis.

Our study makes use of accounting data to address two issues that have appeared in the literature. Firstly, in response to concerns raised by, amongst others, Berk (1995, 1997) that using a company's market capitalization as a measure of firm size may lead to a spurious correlation between size and realized returns, we replicate our test of the size effect using sorts on the principal component of market-based and non-market-based (i.e., "fundamental" or accounting-based) size measures, such as sales and total assets. Secondly, to avoid our results being driven by marginal or financially distressed small firms (e.g., Chan and Chen, 1991), we filter our sample for such firms based on accounting-based measures of profitability and sales. Our results provide evidence of a small but economically significant size effect. In particular, the results indicate that whereas the average share price returns of European firms in the first nine size deciles do not or only moderately vary with firm size, the average returns in the bottom size decile are significantly greater than those in the other deciles, irrespective of the firm size measure used to rank company shares. This finding confirms the presence of a size premium, but only for the smallest firms in the European

sample. Furthermore, we find initial evidence that size premiums in the Nordic countries, Ireland and the United Kingdom exceed those in the other European countries, especially in terms of significance and stability over time.

A study closely related to ours is that of Heston et al. (1999, hereafter HRW), who examine the effect of beta and firm size on European share returns between 1980 and 1995 and find a close to monotonic relationship between firm size and returns. Our study differs from and contributes to HRW in at least two ways. Firstly, we examine a more recent time period, during which economic integration among European countries and exchanges has increased and regulatory interventions aimed at facilitating cross-border investments may have narrowed risk differences between small and large firms. Secondly, because the availability of

return data has improved over time, especially for smaller firms, we can examine a broader range of firm size than HRW. To illustrate, the average natural log of market capitalization in HRW's bottom size decile is close to 3.0, which is slightly greater than the average natural log of market capitalization in our ninth size decile. Interestingly, especially the smallest company shares, being those in the tenth (bottom) size decile of our sample, earn a significant size premium. Only in the top nine deciles we find a close-to-linear relationship between firm size and returns that is economically and statistically significant.

1.2 Background

In one of the first empirical studies on the size effect, Banz (1981) shows that in the period 1926 - 1975 realized returns on US common shares are negatively and nonlinearly related to their market values, suggesting that small cap shares earn a premium. Early studies following Banz (1981) confirm the existence of an average size premium effect and document some of its peculiarities. For example, Brown et al. (1983) show that during their sample period (1967 - 1979) the effect is (1) not stable over time and (2) linear if the log of market value is used. Further, Keim (1983) and Lamoureux and Sanger (1989) find that a substantial proportion of the size premium is earned in January.

During the three decades following Banz (1981), researchers have challenged the existence of the size effect using various arguments. One set of studies argues that firm size is a proxy for underlying risk characteristics, i.e., that small cap and large cap firms' share prices respond differently to changes in risk factors. For example, studies claim that the size effect disappears if one adequately controls for beta risk, measuring betas (1) using annual (rather than daily or monthly) returns (Handa et al., 1989), (2) over longer periods (Chan and Chen, 1988), or (3) using a broader market index including debt and equity claims (Ferguson and Shockley, 2003). Similarly, Chan et al. (1985) show that part of the size premium disappears after controlling for observable macroeconomic risk factors. Furthermore, Chan and Chen (1991) provide evidence that is consistent with the idea that very small firms with low production efficiency and high leverage drive the size effect. As Jegadeesh (1992) points out, however, if firm size measures and observable risk characteristics are closely correlated, empirically disentangling the effects of size and risk on returns becomes a difficult task. In fact, examining size portfolios that have (almost) equal betas, he shows that the size effect cannot be explained by beta risk.

Given that small cap shares typically have lower liquidity and higher transaction costs than large cap shares, some studies test whether the small versus large share price return difference reflects investors' compensation for liquidity risk and/or transaction costs. Stoll and Whaley (1983) find that the size effect disappears after controlling for differences in transaction costs between small and large company shares. However, Schultz (1983) shows that this finding is not robust to a change in sample composition. Other studies indicate that illiquidity (risk) is positively associated with returns (e.g., Amihud and Mendelson, 1986; Brennan and Subrahmanyam, 1996; Datar et al., 1998). Nonetheless, some of these studies explicitly show that firm size remains significant in explaining returns also after controlling for liquidity (risk), suggesting that liquidity differences do not fully explain the size effect (van Dijk, 2011).

Lo and MacKinlay (1990) argue that the observed association between market value and realized returns may be an empirical irregularity uncovered through data snooping, as opposed to a theoretically robust finding. This risk arises because many studies examine overlapping samples of share price returns and are motivated by prior empirical findings rather than theory. To counter this argument some studies analyze the stability of the size effect over time (e.g., Horowitz et al., 2000a, b) or the existence of the size effect on non-US stock exchanges (e.g., Bagella et al., 2000; Beedles, 1992; Chan et al., 1991; Elfakhani et al., 1998; Rouwenhorst, 1999), producing mixed evidence. A limitation of these studies could be, however, that a focus on shorter time periods or smaller non-US samples reduces statistical power and may produce unreliable results. Further, if risk differences between small cap and large cap firms drive the size effect, it is plausible that large cap firms' share returns exceed small cap firms' share returns in some time periods or subsamples (van Dijk, 2011).

In conclusion, prior research on the size effect has produced mixed evidence and has stirred

up much debate on the theoretical and empirical justifications for including a size premium in cost of equity estimates. The fragility of prior evidence leads van Dijk (2011) to conclude that “more empirical research is needed to establish the validity of the size effect in both US and international stock returns.” From a practical point of view, even if firm size proxies for other, potentially unknown or unobservable risk factors, the size effect remains important to researchers

and practitioners as long as firm size is an easily accessible measure of the underlying risk construct. For example, if one were estimating the cost of equity capital for a closely-held firm, then size as measured by accounting-based size metrics may be a valuable proxy for other risk factors. These observations motivate us to re-examine the size effect in a recent large international sample.

1.3 Sample and Data

Following prior research, we test for the presence of a size effect in returns by sorting realized returns on (beginning-of-year) firm size and comparing average annual returns across portfolios. As Berk (1995, 1997) argues, market capitalization and future returns may be correlated not only because of small cap versus large cap firm differences in operating characteristics, but also because high discount rates negatively affect current market value while positively affecting future returns. To examine the influence of such market effects, we sort returns on market capitalization as well as on a composite measure derived from both market-based and accounting-based measures of firm size (described in more detail in section 1.4).

Because we use market and accounting-based data in our analyses, our sample of companies comes from the intersection of the Datastream and Worldscope databases. We focus our analysis on a set of 17 Western European countries (and exchanges) that have exhibited a large degree of integration during the past two decades: Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, and the United Kingdom. Worldscope data (fundamental or accounting-based data) is available for some of our sample companies going back to 1982; however, earlier data consists of backfilled histories for companies that were added to Worldscope in 1987 (Worldscope's year of establishment) or later. In fact, Worldscope data is available for close to 24 percent of the Datastream returns in our initial sample in 1982, increasing to 67 percent in 1990 and 87 percent in 2010. To mitigate selection bias (while preserving statistical power), we therefore start our analysis in 1990.

We exclude from our sample companies that operate in financial service industries because the accounting-based measures of firm size are interpreted differently in those industries. Furthermore, to avoid the potential for the

observed size effect to be primarily driven by highly illiquid shares or firms in the early stages of their life cycle (i.e., "turtle eggs"; see Knez and Ready, 1997), we also exclude:¹

- Firm-year observations that have trading (measured through non-zero returns) on less than five percent of the maximum number of trading days during a calendar year.
- Firm-year observations that lack three years of publicly traded price history or have sales below €1 million in any of the previous three fiscal years.

Finally, to improve the practical relevance of our research and avoid that an observed size effect is primarily driven by marginal firms (Chan and Chen, 1991), we separately examine financially distressed company shares. That is, we design our study to address whether a size premium is present when dealing with a non-start-up, financially healthy company—the typical company in most investment portfolios. Consequently, we filter our sample to identify and separately examine the following firm-years, which we label 'high financial risk' firm years:

- Firm-year observations with negative book equity in any of the previous three fiscal years, and
- Firm-year observations with negative average EBITDA (earnings before interest, taxes, depreciation and amortization), operating profit, or net income in the previous three fiscal years.

We translate market capitalization and all financial statement data from local currencies to euros. To ensure consistency over time, and through the introduction of the euro and the determination of fixed euro conversion rates, we translate all amounts from the perspective of a German investor. This implies that in the years prior to the euro introduction, we translate local currencies to Deutschmarks (DM) first and then from Deutschmarks to euros using the fixed DM-euro exchange rate.²

The share price returns in our analysis are annual buy-and-hold returns, measured as the sum of (1) the annual percentage change in share price plus (2) the dividend yield.³ We calculate share price returns also from the perspective of a German investor. That is, in the years prior to the euro adoption year, we translate all local returns to Deutschmark returns.⁴ To mitigate a potential delisting bias, in calculating rates of return we impute a 30%

loss in the month of delisting for all delistings that are possibly performance-related. We classify a delisting as performance-related when (1) Datastream identifies the reason for delisting as unrelated to a merger or acquisition or as unknown and (2) the firm reported a net loss in the year prior to the delisting.⁵

¹ We acknowledge that the cut-off points used in filtering the sample are unavoidably arbitrary. Our conclusions are, however, robust to moderate changes in cut-offs. Furthermore, we emphasize that because we filter the sample based on information known to investors at the beginning of the return window, the deletion of firms does not introduce selection bias. That is, the exclusion of firms or categorizing them as high-financial risk is based on past financial performance or trading history as of the date that we form the portfolios for any given year and, therefore, not based on any unusual foresight on the part of hypothetical investors in these portfolios.

² The euro was introduced in January 1999. The fixed DM-euro exchange rate is €1 = DM1.95583 (see e.g., <http://www.ecb.int/euro/intro/html/index.en.html>). Our conclusions remain unchanged if we translate local returns into ECU (European Currency Unit) prior to 1999.

³ Using annual buy-and-hold returns (rather than short-window returns) helps to avoid the risk that bid-ask effects as described by Blume and Stambaugh (1983) upwardly bias the estimated size effect. To reduce the influence of data errors in Datastream on our calculation of share price returns, we use the following procedure. Firstly,

we identify observations for which the current annual logarithmic return exceeds 1 and the previous annual logarithmic return is less than -1 (or vice versa). Secondly, assuming that return reversals of such magnitude are the consequence of data errors (see e.g., Ince and Porter, 2006), we replace the returns for these two firm-years with their average value.

⁴ Given that our choice to calculate all returns from the perspective of a German investor is admittedly arbitrary, we replicate the analyses after calculating the returns from the perspective of a UK investor (i.e., translating all local returns into Great Britain Pound returns and using returns on a UK long-term government bond index to calculate the risk-free rate). The results of this sensitivity check indicate that the choice between the German and the UK perspective does not significantly influence our conclusions.

⁵ The imputed 30 percent loss is based on Shumway (1997)'s estimate of average delisting returns. Note that this affects only a small fraction of the firms in our sample. Given the absence of detailed information about delisting (returns) in Datastream, more sophisticated corrections are difficult to make.

1.4 Empirical Results

Descriptive Statistics

Panel A of Table 1.1 displays the effect of the various sample selection criteria on sample size. In the period 1990 - 2018, the intersection of Datastream and Worldscope contains 101,047 annual return observations. The requirement that firms included in the sample are not start-up firms (i.e., firms listed for less than three years or having sales below €1 million in any of the three prior years) reduces the sample size by more than one-fourth (with 29,038 observations). Performance-related selection criteria, aimed at excluding and separately examining marginal or financially distressed firms, reduce the sample size by close to one-fifth (with 19,958 observations). The final (low financial risk) sample contains 50,407 observations, most of which come from Germany, France, and the United Kingdom (see Panel B of Table 1.1).

Table 1.2 presents descriptive statistics for annual buy-and-hold returns and six measures of firm size: market capitalization, market value of invested capital, book value of equity, total assets, sales, and number of employees. The statistics show that there is large sample variation in firm size. To illustrate, the interquartile range of market capitalization is €1,004.1 million (€165.7 million) in the low financial risk (high financial risk) sample. Similarly, the interquartile range of total assets is €1,328.1 million (€382.3 million). The average (median) annual return in the low financial risk sample is 14.5 percent (8.8 percent), which implies an average premium of 8.5 percent over the average risk-free rate of 5.9 percent.⁶ Annual returns in the high financial risk sample are lower, on average, and more volatile than the annual returns in the low financial risk sample. Furthermore, firms in the low financial risk sample are, on average, larger than firms in the high financial risk sample, irrespective of the size measure chosen.

TABLE 1.1: Sample Composition

Panel A: Sample composition

	1990 - 1997	1998 - 2004	2005 - 2011	2012 - 2018	All years
Firm-years with return and accounting data	22,272	28,237	27,702	22,836	101,047
Minus: Firm-years of start-up firms	6,288	9,591	7,902	5,257	29,038
Minus: Firm-years with thin trading	387	609	295	353	1,644
Minus: High-financial-risk firm-years	3,675	5,045	5,937	5,301	19,958
Total number of firm-years in the low-financial-risk sample	11,922	12,992	13,568	11,925	50,407

TABLE 1.1 NOTES: In Panel A, start-up firms are firms that have been listed on a European exchange for less than three years at the start of the calendar year or have sales below €1 million in any of the previous three fiscal years. Firm-years with thin trading are firm-years with non-zero returns on less than 5 percent of all trading days. High financial risk firm-years are firm-years with (a) negative book equity in any of the previous three fiscal years or (b) negative average EBITDA (earnings before interest, taxes, depreciation and amortization), operating profit, or net income in the previous three fiscal years.

TABLE 1.1 - Continued

Panel B: Number of firm-year observations by country and period (in the low financial risk sample)

	1990 - 1997	1998 - 2004	2005 - 2011	2012 - 2018	All years
AUSTRIA	167	195	197	195	754
BELGIUM	256	276	363	322	1,217
DENMARK	386	487	415	385	1,673
FINLAND	117	380	479	401	1,377
FRANCE	1,382	2,024	2,383	2,073	7,862
GERMANY	871	1,317	1,846	1,821	5,855
GREECE	172	688	827	371	2,058
IRELAND	183	157	115	103	558
ITALY	564	458	642	631	2,295
LUXEMBOURG	9	32	41	24	106
NETHERLANDS	668	616	463	321	2,068
NORWAY	228	293	390	377	1,288
PORTUGAL	127	168	113	122	530
SPAIN	312	469	478	344	1,603
SWEDEN	433	592	808	814	2,647
SWITZERLAND	558	663	746	714	2,681
UNITED KINGDOM	5,489	4,177	3,262	2,907	15,835
TOTAL	11,922	12,992	13,568	11,925	50,407

TABLE 1.2: Descriptive statistics of returns and firm size measures

Panel A: Low financial risk sample

	Mean	Median	Std Dev	Q1	Q3
Annual return	0.1446	0.0880	0.5106	-0.1285	0.3426
Currency return	0.0002	0.0000	0.0555	0.0000	0.0000
Risk-free return	0.0594	0.0424	0.0612	0.0152	0.0956
Market capitalization	2,804.6	233.6	12,089.9	60.3	1,064.4
MV of invested capital	3,803.3	320.1	15,365.8	86.4	1,470.4
Book equity	1,236.2	129.7	5,481.1	39.3	525.2
Total assets	3,641.6	309.3	15,428.1	89.8	1,417.9
Sales	2,913.5	331.2	11,954.7	100.9	1,412.2
Employees	11,174	1,574	35,947	481	6,244

TABLE 1.2 - Continued

Panel B: High financial risk sample

	Mean	Median	Std Dev	Q1	Q3
Annual return	0.1148	0.0050	0.7187	-0.2623	0.3313
Currency return	-0.0009	0.0000	0.0464	0.0000	0.0000
Risk-free return	0.0556	0.0424	0.0577	0.0152	0.0774
Market capitalization	659.3	44.3	4,766.2	14.7	180.4
MV of invested capital	1,085.9	82.0	6,924.3	26.4	316.9
Book equity	434.0	34.5	4,657.3	11.3	127.9
Total assets	1,517.4	109.6	9,428.3	34.0	416.3
Sales	1,213.7	100.9	5,709.1	29.0	383.8
Employees	5,150	529	22,178	158	1,982

TABLE 1.2 NOTES: The number of observations is 50,407 and 19,958 in the low financial risk sample and high financial risk sample, respectively. The variables have the following definitions:

Annual return: Annual buy-and-hold return, measured as the sum of (1) the annual percentage change in euro-denominated share price plus (2) dividend yield.

Currency return: Difference between the annual buy-and-hold return in local currency and the annual buy-and-hold return in euros.

Risk-free return: Annual return on the BofA Merrill Lynch German long-term (five year +) government bond index.

Market capitalization: Number of common shares outstanding times share price at the beginning of the calendar year.

MV of invested capital: Market capitalization of equity at the beginning of the calendar year plus the book value of debt at the end of the most recent fiscal year prior to the calendar year.

Book equity: Book value of equity at the end of the most recent fiscal year prior to the calendar year.

Total assets: Book value of total assets at the end of the most recent fiscal year prior to the calendar year.

Sales: Total revenues of the most recent fiscal year prior to the calendar year.

Employees: Number of employees at the end of the most recent fiscal year prior to the calendar year.

All monetary amounts are in € millions and inflation-adjusted using the German Consumer Price Index.

Market Capitalization Portfolios

Panel A of Table 1.3 displays the arithmetic averages of annual equal-weighted returns of ten firm size portfolios. To create ten equal-sized portfolios we sort all shares in the full sample by market capitalization at the beginning of the calendar year. We use the breakpoints determined in the full sample also to create firm size portfolios in the low financial risk and high financial risk samples.⁷

Because financial risk is negatively correlated with firm size, as suggested by, for example, Chan and Chen (1991), the bottom size portfolio contains relatively few (many) observations in the low financial risk (high financial risk) sample. In particular, at the end of the sample period, in 2018, the bottom size portfolio in the full sample contains 233 firm year observations, of which significantly less than half (31.3 percent) is assigned to the low financial risk sample.

We choose to report equal-weighted rather than value-weighted returns because we consider the former type of returns most relevant to the analyst applying the results of our study in practice. In particular, our analysis of equal-weighted returns mimics the approach of an analyst matching his or her subject firm with a group of comparable firms based on a set of risk characteristics that includes firm size and financial risk. As the

returns of each firm in the group of comparable firms are potentially equally informative about the effect of matching (risk) characteristics on the cost of equity, relevant to the analyst are the equal-weighted realized returns in the comparables group, not the value-weighted returns. Using equal-weighted returns in our study is also analogous to analysts using equal-weighted valuation multiples in market comparables valuation methods.

TABLE 1.3: Average Annual Returns and Premiums of Market Capitalization Portfolios

Panel A: Equal-weighted returns

Full sample					
Size portfolio	Upper bound in 2018	N in 2018	Premium over r_f	Portfolio beta	Premium over CAPM
1 (big)	237,272.4	233	4.48%	0.94	0.00%
2	7,712.5	234	6.57%	1.07	1.47%
3	2,603.7	233	5.89%	1.09	0.70%
4	1,167.5	234	6.17%	1.07	1.07%
5	567.5	233	6.45%	1.06	1.40%
6	292.4	234	6.90%	1.04	1.94%
7	146.8	234	6.73%	1.02	1.87%
8	78.6	233	6.43%	1.06	1.38%
9	38.1	234	7.21%	1.11	1.92%
10 (small)	15.3	233	13.25%	1.17	7.67%
Pooled	–	2,335	7.01%	1.06	1.96%

Low-financial-risk sample					
Size portfolio	Upper bound in 2018	N in 2018	Premium over r_f	Portfolio beta	Premium over CAPM
1 (big)	237,272.4	215	4.75%	0.93	0.32%
2	7,712.5	214	7.01%	1.05	2.01%
3	2,603.7	206	6.31%	1.08	1.16%
4	1,167.5	191	7.20%	1.04	2.24%
5	567.5	200	7.59%	1.01	2.78%
6	292.4	176	8.47%	0.99	3.75%
7	146.8	159	9.35%	0.96	4.77%
8	78.6	137	9.22%	1.00	4.45%
9	38.1	106	10.86%	1.02	6.00%
10 (small)	15.3	73	18.36%	1.22	12.55%
Pooled	–	1,677	8.17%	1.02	3.31%

TABLE 1.3, Panel A - Continued

Size portfolio	Upper bound in 2018	High-financial-risk sample			
		N in 2018	Premium over r_f	Portfolio beta	Premium over CAPM
1 (big)	237,272.4	18	2.64%	1.11	(2.65%)
2	7,712.5	20	4.07%	1.17	(1.51%)
3	2,603.7	27	3.55%	1.14	(1.88%)
4	1,167.5	43	0.73%	1.19	(4.94%)
5	567.5	33	2.15%	1.21	(3.62%)
6	292.4	58	2.53%	1.17	(3.05%)
7	146.8	75	1.41%	1.17	(4.17%)
8	78.6	96	2.29%	1.13	(3.10%)
9	38.1	128	3.26%	1.20	(2.46%)
10 (small)	15.3	160	9.82%	1.13	4.43%
Pooled	–	658	3.90%	1.15	(1.58%)

TABLE 1.3 NOTES: This table displays average annual excess returns over the risk-free rate of portfolios sorted on market capitalization. Portfolios in the full sample are equal-sized. Breakpoints determined in the full sample have been used to create portfolios in the other samples. Portfolio betas are post-ranking betas calculated as the sum of the slopes in a regression of monthly returns on contemporaneous and once-lagged index returns. Premiums over CAPM are calculated as excess returns minus beta times the market risk premium, where the market risk premium has been set equal to the excess return in the top size portfolio of the full sample.

The average excess return over the risk-free rate is 7.01 percent in the full sample. We calculate average portfolio returns by first averaging returns (equal-weighted) across shares within a year and then averaging annual portfolio returns (equal-weighted) across years. This procedure ensures that later years in the sample, which have relatively many observations, do not receive more weight than earlier years.⁸ Average excess returns increase, albeit non-monotonically, from 4.48 percent in the top portfolio to 13.25 percent in the bottom portfolio, thus providing initial support for the idea that the firm size effect is also present in the European sample. The relationship between firm size and excess returns is evidently non-linear, both in the full sample and in the low financial risk sample. We

explore this finding in more detail later in this report.

To control for the possibility that marginal, financially distressed firms in the sample drive the observed firm size effect we also calculate average excess returns for low financial risk and high financial risk firm-years separately. In the low financial risk sample the excess return differential between the top and bottom portfolios is 13.61 percent (versus 8.77 percent in the full sample), indicating that the effect of firm size on excess returns in the European sample is not—or at least not fully—attributable to financial risk. This top-bottom portfolio return spread is statistically significant at the 1 percent significance level based on a (non-tabulated) standard t-test that uses the standard deviation in 29 annual return differences to calculate standard errors.⁹

Table 1.3 also reports post-ranking portfolio betas, which we calculate as the sum of the slopes in a regression of monthly portfolio returns on contemporaneous and lagged monthly MSCI Europe index returns (cf. Fama and French, 1992). Consistent with prior US evidence, portfolio betas decrease with firm size. Because regressions of portfolio returns on firm size and beta may not accurately separate the beta effect from the firm size effect (see e.g., Jegadeesh, 1992), we use three alternative methods to control for beta. A first method is to calculate premiums over CAPM as the difference between the average equal-weighted excess portfolio return and beta times the market risk premium, which we set equal to the (beta-adjusted) excess return of the top portfolio (i.e., 4.48/0.94, or 4.77 percent in Panel A of Table 1.3). For the analyst using the results of this study to estimate a firm's cost of equity, the cost of equity size premium is best approximated by the premium over CAPM measure.

The results in Table 1.3 show that patterns of premiums over CAPM are consistent with patterns of excess returns, suggesting that beta may reduce but not explain the size premium. As a second way to control for beta, in Panel B of Table 1.3 we assign individual

weights (w_i) to each of the n firm-year specific returns in a size portfolio. These weights solve the following quadratic optimization problem (cf. Jegadeesh, 1992):

$$\begin{aligned} &\text{Minimize } \sum_{i=1}^n w_i^2 \\ &\text{Subject to } \sum_{i=1}^n w_i \beta_i = \text{target } \beta \\ &\quad \text{and } \sum_{i=1}^n w_i = 1 \end{aligned}$$

where β_i is the beta of firm-year i and the target β has been set equal to one, such that the weighted average beta of each size portfolio approaches the market beta.¹⁰

Panel B of Table 1.3 presents the beta risk-weighted (w_i -weighted) excess portfolio returns. The evidence in Panel B confirms our previous finding that the firm size effect is present in European share price returns, also after controlling for beta differences between size portfolios. In particular, in the full sample the average excess return increases from 4.39 percent in the top size portfolio to 13.14 percent in the bottom size portfolio. Furthermore, in the low financial risk sample the return difference between the top and bottom size portfolios equals 11.83 percent and is statistically significant at the 1 percent significance level.¹¹

TABLE 1.3: Average Annual Returns and Premiums of Market Capitalization Portfolios

Panel B: Beta-weighted returns

Size portfolio	Upper bound in 2018	Full sample			
		N in 2018	Premium over rf	Portfolio beta	Premium over CAPM
1 (big)	237,272.4	233	4.39%	1.00	0.00%
2	7,712.5	234	6.78%	1.00	2.39%
3	2,603.7	233	5.81%	1.01	1.38%
4	1,167.5	234	6.23%	1.01	1.80%
5	567.5	233	6.37%	1.01	1.94%
6	292.4	234	6.75%	1.01	2.32%
7	146.8	234	6.80%	1.01	2.37%
8	78.6	233	6.52%	1.01	2.09%
9	38.1	234	7.66%	1.01	3.23%
10 (small)	15.3	233	13.14%	1.02	8.66%
Pooled	—	2,335	7.04%	1.01	2.61%

A third way to control for beta risk differences across size portfolios is to create beta-size portfolios and regress portfolio returns on portfolio averages of firm size and beta (cf.

Heston et al., 1999). We discuss this regression in a later section of this report.

TABLE 1.3, Panel B - Continued

Low-financial-risk sample					
Size portfolio	Upper bound in 2018	N in 2018	Premium over rf	Portfolio beta	Premium over CAPM
1 (big)	237,272.4	215	4.61%	1.00	0.22%
2	7,712.5	214	7.14%	1.00	2.75%
3	2,603.7	206	6.22%	1.00	1.83%
4	1,167.5	191	7.18%	1.01	2.75%
5	567.5	200	7.59%	1.01	3.16%
6	292.4	176	8.55%	1.01	4.12%
7	146.8	159	9.19%	1.01	4.76%
8	78.6	137	9.26%	1.01	4.83%
9	38.1	106	10.96%	1.01	6.53%
10 (small)	15.3	73	16.44%	1.03	11.92%
Pooled	–	1,677	8.71%	1.01	4.28%
High-financial-risk sample					
Size portfolio	Upper bound in 2018	N in 2018	Premium over rf	Portfolio beta	Premium over CAPM
1 (big)	237,272.4	18	2.64%	1.02	(1.84%)
2	7,712.5	20	5.81%	1.01	1.38%
3	2,603.7	27	3.65%	1.04	(0.92%)
4	1,167.5	43	0.99%	1.02	(3.49%)
5	567.5	33	1.73%	1.04	(2.84%)
6	292.4	58	0.80%	1.01	(3.63%)
7	146.8	75	2.00%	1.03	(2.52%)
8	78.6	96	2.57%	1.01	(1.86%)
9	38.1	128	2.70%	1.00	(1.69%)
10 (small)	15.3	160	10.07%	1.04	5.50%
Pooled	–	658	3.30%	1.02	(1.18%)

TABLE 1.3 NOTES: This table displays average annual excess returns over the risk-free rate of portfolios sorted on market capitalization. Portfolios in the full sample are equal-sized. Breakpoints determined in the full sample have been used to create portfolios in the other samples. Portfolio betas are post-ranking betas calculated as the sum of the slopes in a regression of monthly returns on contemporaneous and once-lagged index returns. Premiums over CAPM are calculated as excess returns minus beta times the market risk premium, where the market risk premium has been set equal to the excess return in the top size portfolio of the full sample.

Size Factor Portfolios

Prior research has suggested that the association between beginning-of-year market capitalization and subsequent returns may be spurious because market capitalization depends on investors' discount rate assumptions (see e.g., Berk, 1995, 1997). In response to this concern, we construct portfolios using an alternative measure of firm size. The idea underlying this alternative measure is that the principal component of market capitalization and non-market variables, such the book value of assets or sales, accurately reflects firm size without being dependent on investors' discount rate. A principal component analysis of the annual (beginning-of-year) values of six firm size measures produces the following factor score equation:¹²

$$\begin{aligned} \text{Factor} = & -2.882 + \\ & 0.080 \times \ln(\text{Market capitalization}) + \\ & 0.084 \times \ln(\text{MV of invested capital}) + \\ & 0.087 \times \ln(\text{Book equity}) + \\ & 0.088 \times \ln(\text{Total assets}) + \\ & 0.087 \times \ln(\text{Sales}) + \\ & 0.084 \times \ln(\text{Employees}) \end{aligned}$$

where all variables are as defined in Table 1.2.

Table 1.4 displays the arithmetic averages of the annual equal-weighted (Panel A) and beta

risk-weighted (Panel B) excess returns of ten size factor portfolios. Like in Table 1.3, we sort all shares in the full sample by size factor at the beginning of the calendar year and use the breakpoints of the full sample to create size portfolios in the low financial risk and high financial risk samples. In the full sample (low financial risk sample) average equal-weighted excess returns increase non-monotonically from 5.15 percent (5.25 percent) in the top portfolio to 11.37 percent (14.92 percent) in the bottom portfolio, confirming our prior finding that the smallest firms in the sample earn a premium relative to the largest firms. In both the full sample and the low financial risk sample the top versus bottom portfolio difference in premium over CAPM is also positive (4.82 and 7.62 percent, respectively). Furthermore, in Panel B, in which we calculate beta risk-weighted portfolio returns to control for the effect of firm size on beta, average excess returns in the bottom size factor portfolio are significantly greater than those in the top portfolio (11.12 versus 5.15 percent in the full sample; 13.56 versus 5.20 percent in the low financial risk sample). In summary, these findings indicate that our conclusion that the firm size effect on returns is present in Europe is robust to changes in how we measure firm size.

TABLE 1.4: Average Annual Returns and Premiums of Size Factor Portfolios

Panel A: Equal-weighted returns

Size portfolio	Upper bound in 2018	Full sample			
		N in 2018	Premium over rf	Portfolio beta	Premium over CAPM
1 (big)	3.42	233	5.15%	1.00	0.00%
2	1.64	234	7.46%	1.08	1.90%
3	1.12	233	6.56%	1.07	1.05%
4	0.71	234	5.73%	1.07	0.22%
5	0.36	233	5.35%	1.06	(0.11%)
6	0.07	234	6.04%	1.03	0.74%
7	(0.22)	234	6.88%	1.05	1.47%
8	(0.56)	233	7.22%	0.99	2.12%
9	(0.85)	234	8.31%	1.08	2.75%
10 (small)	(1.27)	233	11.37%	1.17	5.34%
Pooled	–	2,335	7.01%	1.06	1.55%

TABLE 1.4, Panel A – Continued

Low-financial-risk sample					
Size portfolio	Upper bound in 2018	N in 2018	Premium over rf	Portfolio beta	Premium over CAPM
1 (big)	3.42	208	5.25%	0.97	0.25%
2	1.64	213	7.50%	1.06	2.04%
3	1.12	203	7.24%	1.06	1.78%
4	0.71	199	7.04%	1.05	1.63%
5	0.36	183	6.35%	1.02	1.10%
6	0.07	185	8.20%	0.99	3.10%
7	(0.22)	157	8.86%	0.99	3.76%
8	(0.56)	147	9.52%	0.92	4.78%
9	(0.85)	114	11.34%	1.00	6.19%
10 (small)	(1.27)	68	14.92%	1.22	8.64%
Pooled	–	1,677	8.17%	1.02	2.92%

High-financial-risk sample					
Size portfolio	Upper bound in 2018	N in 2018	Premium over rf	Portfolio beta	Premium over CAPM
1 (big)	3.42	25	4.21%	1.27	(2.33%)
2	1.64	21	7.21%	1.19	1.08%
3	1.12	30	2.35%	1.16	(3.62%)
4	0.71	35	0.18%	1.18	(5.90%)
5	0.36	50	1.39%	1.22	(4.89%)
6	0.07	49	1.03%	1.13	(4.79%)
7	(0.22)	77	2.30%	1.19	(3.83%)
8	(0.56)	86	3.46%	1.11	(2.26%)
9	(0.85)	120	4.28%	1.18	(1.80%)
10 (small)	(1.27)	165	8.55%	1.16	2.58%
Pooled	–	658	3.90%	1.15	(2.02%)

TABLE 1.4 NOTES: This table displays average annual excess returns over the risk-free rate of portfolios sorted by size. Portfolios in the full sample are equal-sized. Breakpoints are determined using the factor score of a principal component analysis of the natural logs of the following firm size measures: market capitalization, market value of invested capital, book value of equity, book value of total assets, sales, and number of employees (as defined in Table 1.2). Breakpoints determined in the full sample are used to create portfolios in the other samples. Portfolio betas are post-ranking betas calculated as the sum of the slopes in a regression of monthly returns on contemporaneous and once-lagged index returns. Premiums over CAPM are calculated as excess returns minus beta times the market risk premium, where the market risk premium has been set equal to the excess return in the top size portfolio of the full sample.

TABLE 1.4 – Continued

Panel B: Beta-weighted returns

Size portfolio	Upper bound in 2018	Full sample			
		N in 2018	Premium over rf	Portfolio beta	Premium over CAPM
1 (big)	3.42	233	5.15%	1.00	0.00%
2	1.64	234	7.31%	1.00	2.16%
3	1.12	233	6.79%	1.00	1.64%
4	0.71	234	5.58%	1.01	0.38%
5	0.36	233	5.41%	1.01	0.21%
6	0.07	234	6.14%	1.01	0.94%
7	(0.22)	234	6.88%	1.01	1.68%
8	(0.56)	233	7.12%	1.02	1.87%
9	(0.85)	234	8.61%	1.01	3.41%
10 (small)	(1.27)	233	11.12%	1.02	5.87%
Pooled	–	2,335	7.01%	1.01	1.81%
Size portfolio	Upper bound in 2018	Low-financial-risk sample			
		N in 2018	Premium over rf	Portfolio beta	Premium over CAPM
1 (big)	3.42	208	5.20%	1.00	0.05%
2	1.64	213	7.44%	1.00	2.29%
3	1.12	203	7.41%	1.00	2.26%
4	0.71	199	6.92%	1.01	1.72%
5	0.36	183	6.36%	1.01	1.16%
6	0.07	185	8.14%	1.01	2.94%
7	(0.22)	157	8.97%	1.01	3.77%
8	(0.56)	147	9.49%	1.02	4.24%
9	(0.85)	114	11.35%	1.01	6.15%
10 (small)	(1.27)	68	13.56%	1.01	8.36%
Pooled	–	1,677	8.48%	1.01	3.28%
Size portfolio	Upper bound in 2018	High-financial-risk sample			
		N in 2018	Premium over rf	Portfolio beta	Premium over CAPM
1 (big)	3.42	25	6.99%	1.01	1.79%
2	1.64	21	5.86%	0.98	0.81%
3	1.12	30	3.38%	1.03	(1.92%)
4	0.71	35	(0.46%)	1.00	(5.61%)
5	0.36	50	0.79%	1.04	(4.57%)
6	0.07	49	1.17%	1.02	(4.08%)
7	(0.22)	77	2.38%	1.00	(2.77%)
8	(0.56)	86	3.58%	1.03	(1.72%)
9	(0.85)	120	4.28%	1.02	(0.97%)
10 (small)	(1.27)	165	8.52%	1.05	3.11%
Pooled	–	658	3.65%	1.01	(1.55%)

TABLE 1.4 NOTES: See notes of Panel A

The results in both Table 1.3 and Table 1.4 suggest that the empirical relationship between firm size and returns is non-linear. To further explore this non-linearity, we perform a regression analysis that is similar to the analysis of HRW (1999). This analysis also allows us to (a) examine the effect of beta risk on portfolio returns, (b) assess the statistical significance of size premiums, and (c) qualitatively compare our results to the results of HRW. In this analysis we first sort shares within each size portfolio on pre-ranking beta into ten groups, thereby constructing 100 beta-size portfolios. We then calculate post-ranking betas, average firm size and average annual excess returns for each beta-size portfolio and perform a Fama-MacBeth regression of annual excess portfolio returns on beta and size, both before and after including indicator (dummy) variables for the bottom size portfolios.

Table 1.5 reports the average coefficients and Fama-MacBeth t-statistics of 29 annual regressions, which we estimate using two different measures of firm size, i.e., market capitalization and size factor, and for the full and low financial risk samples separately. In

Panel A, the regression results in columns 1 and 4 indicate that the relation between firm size and returns is negative and statistically significant in the full and low financial risk samples. After including an indicator (dummy) variable that equals one for observations in the bottom size portfolio (portfolio 10), the coefficient on size in column 5 of Panel A decreases from -1.40 to -0.87 but remains statistically significant at the 1 percent significance level. The coefficient on size further decreases in magnitude and statistical significance after including an indicator (dummy) variable that equals one for observations in the ninth size decile (see column 6 of Panel A). The coefficient on the indicator variable for observations in the bottom decile is positive and statistically significant at the 1 percent significance level, which, combined with the decreased significance of the coefficient on firm size, confirms that the relationship between firm size and European company share returns is strongly non-linear. The close-to-linear relationship between firm size and share returns in the top nine size deciles is not only statistically but also economically significant. In fact, the coefficient on size of -0.87,

TABLE 1.5: Regression Analysis of the Relationship between Excess Portfolio Returns, Firm Size and Beta
Full sample Low financial risk sample

Panel A: Market capitalization portfolios

	Full sample			Low-financial-risk sample		
	1	2	3	4	5	6
Portfolio beta	-0.1531 (-0.046)	-0.4323 (-0.131)	-0.4332 (-0.131)	3.5887 (1.006)	2.5687 (0.721)	2.555 (0.716)
log(Market cap)	-0.7733 (-2.202)	-0.3240 (-0.991)	-0.2977 (-0.933)	-1.3998 (-3.643)	-0.8709 (-2.827)	-0.7704 (-2.372)
Portfolio 9			0.0854 (0.075)			1.0033 (0.793)
Portfolio 10		5.7494 (3.326)	5.7878 (3.139)		6.9066 (2.815)	7.3169 (2.765)

TABLE 1.5 – Continued

Panel B: Size factor portfolios

	Full sample			Low-financial-risk sample		
	1	2	3	4	5	6
Portfolio beta	-0.2994 (-0.089)	-0.7100 (-0.210)	-0.7837 (-0.231)	2.4909 (0.766)	1.7051 (0.510)	1.6433 (0.490)
Size factor	-1.2804 (-1.826)	-0.6360 (-0.897)	-0.3205 (-0.566)	-2.3475 (-3.275)	-1.6825 (-2.576)	-1.2868 (-2.104)
Portfolio 9			1.5503 (0.952)			2.0411 (1.383)
Portfolio 10		3.8421 (2.394)	4.4915 (2.549)		3.8445 (1.823)	4.6926 (1.980)

TABLE 1.5 NOTES: This table displays the average coefficients and Fama-MacBeth t-statistics of 29 annual regressions of excess beta-size portfolio returns on post-ranking portfolio betas, portfolio averages of firm size, and indicator variables for the two bottom size portfolios, for the years 1990 - 2018. In particular:

- Columns 1 and 4 refer to the regression of excess beta-size portfolio returns on portfolio betas and the natural log of market cap (in Panel A) or Size Factor (Panel B)
 - Columns 2 and 5 refer to the regression including a dummy variable for Portfolio 10
 - Columns 3 and 6 refer to the regression including a dummy variable for Portfolios 9 and 10
- Coefficients displayed in bold are statistically significant at (at least) the 10 percent significance level. T-statistics are shown in parentheses.

displayed in column 5, predicts an average return difference of 5.7 percent between the median company in the top size decile (with a log of market capitalization of 9.7 in 2018) and the median company in the ninth size decile (with a log of market capitalization of 3.2 in 2018).

The regression results for the size factor portfolios, displayed in Panel B, are strongly comparable to the results in Panel A.

Country differences

Prior country-specific studies on the firm size effect have produced mixed evidence, leading some researchers and practitioners to conclude that the effect does not exist in some non-US countries.¹³ To shed some light on differences in the significance of the size

effect across economic regions and their potential origin, we split the low financial risk sample into clusters of geographically proximate and economically integrated countries and estimate cluster-specific size premiums.¹⁴ In this analysis we sort shares into four (instead of ten) equal-sized portfolios to preserve statistical power. Furthermore, we determine firm size breakpoints in the full European sample first and then use these breakpoints to construct portfolios in the regional samples, thus ensuring that size portfolios are consistently defined across samples. Finally, to control for beta risk differences across portfolios, we focus the analysis on beta risk-weighted portfolio returns, which we calculate using the quadratic optimization procedure described earlier.

FIGURE 1.1: Average Annual Return Spreads between Top and Bottom Market Capitalization Quartiles by Country or Region

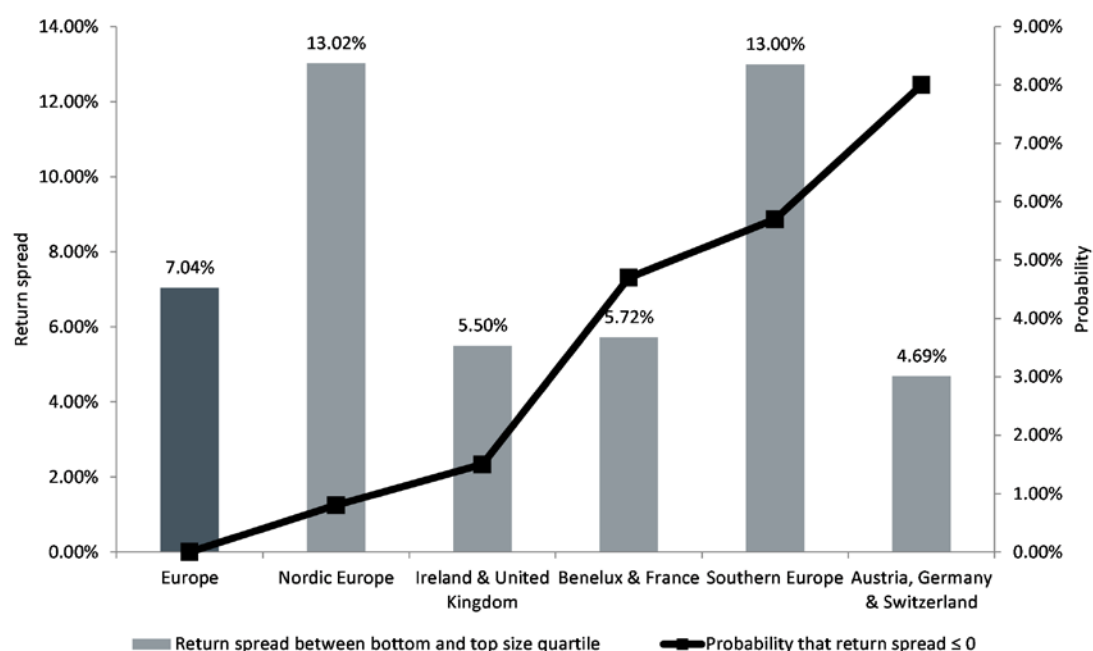


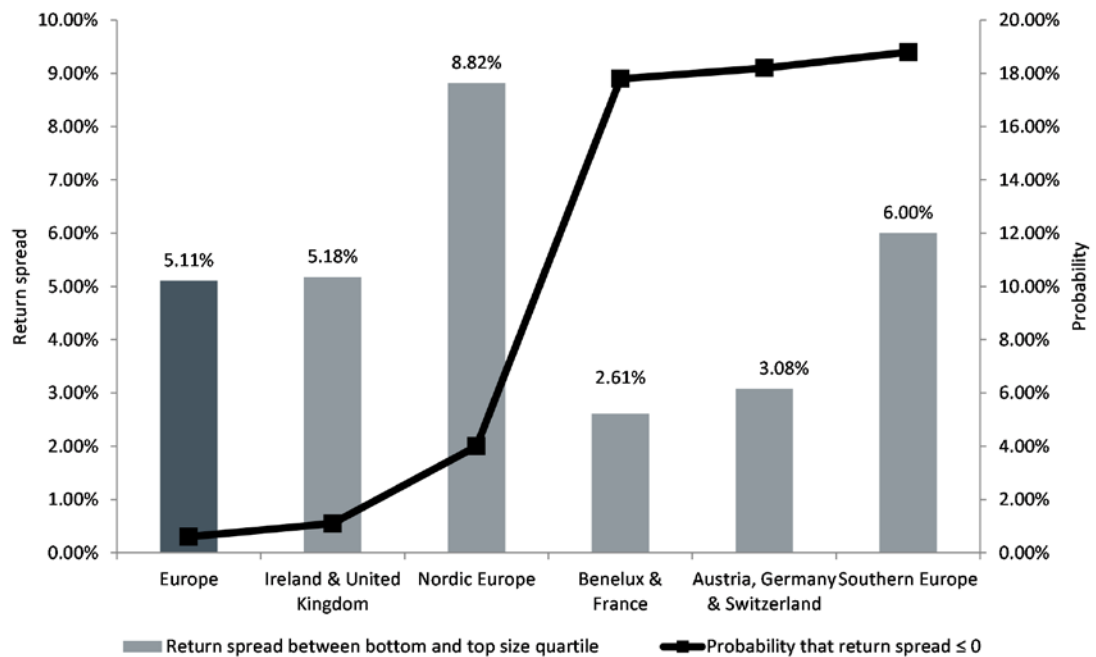
Figure 1.1 displays the average portfolio return spreads between the bottom and top market capitalization quartiles for Europe and the following five regions: (1) Austria, Germany, and Switzerland, (2) Benelux (Belgium, Netherlands, and Luxembourg) and France, (3) Ireland and the United Kingdom, (4) Nordic Europe (Denmark, Finland, Norway, and Sweden), and (5) Southern Europe (Greece, Italy, Spain, and Portugal).¹⁵ The regional return spreads are presented in order of significance, measured as the (one-sided) probability that the differential is positive.¹⁶ Although we caution the reader to interpret the observed regional differences with care, especially because splitting up the sample unavoidably affects statistical power, the pattern arising provides interesting exploratory evidence about how country factors affect the size premium. In particular, Figure 1 shows that the size premium is positive in every economic region considered. However, the magnitude and statistical significance of the return spreads varies considerably. Return spreads are have the

highest (statistical) significance in the Anglo-Saxon (Ireland and the United Kingdom) and Nordic (Denmark, Finland, Norway, and Sweden). In contrast, the return spreads in the other two regions are not significantly different from zero.

Figure 1.2 shows the average portfolio return spreads between the bottom and top quartiles sorted on size factor, i.e., the principal component of six firm size measures. In this Figure, return spreads and probabilities have been calculated in the same way as in Figure 1.1 and return spreads are presented in order of significance. The return spreads displayed in Figure 1.2 confirm that the significance of the size effect varies across European regions. When using size factors to construct portfolios, the return spreads between the bottom and top portfolios are also significantly different from zero in the Anglo-Saxon and Nordic countries, respectively at the 5 percent significance level. The return spreads in the other three regions are positive, though not significantly

different from zero at traditional significance levels.

FIGURE 1.2: Average Annual Return Spreads between Top and Bottom Size Factor Quartiles by Country or Region



The evidence in Figures 1.1 and 1.2 suggests a distinction between Anglo-Saxon and Nordic countries with significant and relatively stable size premiums and Benelux, Germanic and Southern European countries with smaller or more volatile size premiums.¹⁷ Given prior studies on determinants of expected returns, a few potential explanations can be offered for this distinction. Firstly, prior research has indicated that the firm value discount associated with corporate diversification increases with the development of external capital markets (see e.g., Fauver et al., 2003). Consequently, corporate diversification has traditionally been widespread and considered beneficial in bank-oriented economies such as those in the Benelux, Germanic and Southern European regions. During the past two decades public equity markets in these economies have rapidly developed, thus potentially reducing the net benefits of diversification. If during this period the unwinding of corporate conglomerates lagged behind the development of public equity markets, temporary increases in (typically large) conglomerates' required returns may have narrowed the return spread between the top and bottom size portfolios in our sample.

Secondly, firm size distributions differ across the economic regions. To illustrate, recall our earlier finding, displayed in Table 1.5, that very small (low financial risk) firms, belonging to the smallest 10 percent of all European firms, have a relatively strong influence on the firm size-return relationship. In the Anglo-Saxon and Nordic countries (combined) 34.7 percent of the observations in the bottom market capitalization quartile are from such very small firms. In contrast, in the other regions (combined) only 27.3 percent of the observations in the bottom market capitalization quartile are from very small firms. The relatively high percentage of very small firms in the bottom size quartile of the Anglo-Saxon and Nordic countries may positively affect the return differential spreads in these countries displayed in Figures 1.1 and 1.2.

Finally, the relationship between firm size and potential determinants of returns, such as (idiosyncratic) operating risk, financial leverage, share liquidity and default risk, may differ across regions. Table 1.6 reports the mean spreads in such determinants between the top and bottom size quartiles of the two regional clusters. It appears unlikely that firm size-related differences in leverage and operating risk explain why return spreads in

TABLE 1.6: Differences between the Bottom and Top Firm Size Quartiles by Country Cluster

Panel A: Market capitalization portfolios

Variable	Cluster 1		Cluster 2	
	Mean	Standard error	Mean	Standard error
Leverage spread	0.0277	0.0064	0.0765	0.0066
Operating risk spread	0.0151	0.0017	0.0176	0.0022
Liquidity spread	0.5346	0.0085	0.3506	0.0092
Default risk spread	-0.0738	0.0669	-0.1298	0.0527

TABLE 1.6 – Continued

Panel B: Size factor portfolios

Variable	Cluster 1		Cluster 2	
	Mean	Standard error	Mean	Standard error
Leverage spread	-0.0783	0.0051	-0.0705	0.0072
Operating risk spread	0.0258	0.0023	0.0293	0.0020
Liquidity spread	0.5151	0.0095	0.3315	0.0088
Default risk spread	1.3063	0.1081	0.9907	0.0466

TABLE 1.6 NOTES:

Leverage spread: Bottom versus top size quartile difference in the average ratio of the book value of debt to the market value of invested capital.

Operating risk spread: Bottom versus top size quartile difference in average operating risk, measured as the standard deviation in operating margin during the three most recent years prior to the calendar year.

Liquidity spread: Bottom versus top size quartile difference in average share liquidity, measured as the percentage of non-zero return days during the most recent year prior to the calendar year.

Default risk spread: Bottom versus top size quartile difference in average default risk, measured as the Altman z-score at the end of the most recent fiscal year prior to the calendar year.

Mean (standard errors of) spreads: calculated as the averages (standard error) of 29 annual spreads during the sample period 1990 - 2018.

Country cluster 1: consists of the Anglo-Saxon and Nordic countries.

Country cluster 2: consists of the Benelux, Germanic and Southern European countries.

Anglo-Saxon and Nordic countries ("Cluster 1" in Table 1.6) exceed those in the other countries ("Cluster 2" in Table 1.6). In fact, the bottom versus top quartile spreads in leverage and operating risk are greater (or less negative) in the Benelux, Germanic and Southern European countries than in the Anglo-Saxon and Nordic countries. However, liquidity differences between bottom quartile and top quartile observations are significantly smaller in the former regions. Given prior research findings that liquidity may be priced,

liquidity differences between small and large firms might thus explain the observed pattern in return spreads across European countries.

In summary, we find evidence, albeit exploratory, that the size premium varies across countries. Potential causes for the observed country differences in size premiums may be differences in the relationship between firm size and liquidity, size distributions, or corporate diversification.

⁶ In Table 1.2 the average risk-free return in the low financial risk sample is different from that in the high financial risk sample. We obtain this result because the 50,407 and 19,958 observations from, respectively, the low financial risk sample and high financial risk sample are not similarly distributed across years in the sample period. To illustrate, consider the following simplified example in which the risk-free returns in years 1 and 2 are, respectively, 8 percent and 6 percent. If 40 percent of the observations in sample A come from year 1 (i.e., 60 percent from year 2), the average risk-free return in sample A equals 6.8 percent ($0.4 \times 8 + 0.6 \times 6$). If, in contrast, 50 percent of the observations in sample B come from year 1 (i.e., 50 percent from year 2), the average risk-free return in sample B equals 7 percent ($0.5 \times 8 + 0.5 \times 6$). Differences in the distribution of observations across years thus cause the average risk-free returns to differ between the two samples.

⁷ We form portfolios from independent sorts on financial risk and firm size to preserve the variation in firm size in both the low financial risk and high financial risk samples.

⁸ Note that this procedure also implies that the pooled-sample excess returns displayed in Table 1.3 may slightly deviate from excess returns displayed in Table 1.2, where all firm-year observations receive equal weight.

⁹ The observation that excess returns of high financial risk firms are smaller, on average, than those of low financial risk firms goes against the idea that riskier firms should generate higher returns but is consistent with recent evidence demonstrating a negative relationship between default risk and realized share price returns in US stock markets after 1980 (e.g., Campbell et al., 2008; Chava and Purnanandam, 2010).

¹⁰ We estimate firm-year specific betas as follows. Firstly, we calculate pre-ranking betas as the sum of the slopes in a regression of monthly share price returns on contemporaneous and lagged monthly index returns during a period of at least 36 and at most 60 months prior to the start of the calendar year. Secondly, within each year and size portfolio we sort shares on pre-ranking beta to create ten equal-sized beta portfolios. Finally, we estimate post-ranking portfolio betas for each beta-size portfolio and assign the portfolio beta estimates to each firm-year in the portfolio.

¹¹ To test whether the results are robust to changes in beta estimation, we replicate our tests

using annual post-ranking betas. We estimate annual betas by regressing annual portfolio returns on contemporaneous index returns, both measured over rolling overlapping windows of 12 months. This analysis shows that the top versus bottom portfolio difference in annual beta is greater than its monthly equivalent, irrespective of the firm size measure used, which is consistent with the findings of Handa et al. (1989) in a US setting. However, if we use the annual beta to calculate premiums over CAPM or if we assign weights to individual share price returns that solve the quadratic optimization problem such that all annual portfolio betas approach the market beta, our conclusions remain unchanged.

¹² The principal components analysis yields one factor with an eigenvalue greater than one.

¹³ For example, some studies of the German equity market conclude that in recent decades share price returns of small German firms have not significantly exceeded share price returns of large German firms (see e.g., Schulz, 2009). With their view seemingly supported by these findings, the German Institute of Public Auditors ('Institut der Wirtschaftsprüfer') recommends in its (nationally authoritative) Principles for the Performance of Business Valuations not to add size premiums to cost of capital estimates.

¹⁴ If we perform the same analysis on the full (rather than the low financial risk) sample, our conclusions remain qualitatively similar.

¹⁵ Note that because we use beta risk-weighted portfolio returns to calculate return spreads the displayed spreads in excess returns are equal to the (not separately reported) spreads in premiums over CAPM.

¹⁶ We estimate the standard error of return differentials based on 29 annual return differentials (measured during the period 1990 through 2018).

¹⁷ We cannot fully rule out the possibility that the lack of significance observed in three regions is driven by a small sample size in some of the quartiles. As of 2018, the number of firms in the bottom market capitalization quartile ranged from a low of 32 in Austria, Germany, and Switzerland to a high of 61 in Ireland and the United Kingdom. The number of firms in the bottom size factor quartile ranged from 35 in the Germanic regions to 63 in Ireland and the United Kingdom. We caution the reader to interpret the observed regional differences with care.

1.5 Conclusion

This study examines whether the realized share price returns of small European firms have exceeded those of large firms during the period 1990 - 2018. Using various measures of firm size, we find that small company shares have outperformed large company shares, on average, suggesting that investors perceive small European firms as more risky and thus demand a size premium. The evidence also indicates that the relationship between firm size and returns is strongly non-linear.

Breaking the European sample into regional subsamples, we find that the relationship between firm size and returns varies across regions. In particular, we find that the firm size effect is strongest in the Anglo-Saxon and

Nordic countries in the sample. While underlining the potential relevance of this finding, we caution the reader not to automatically conclude from it that firm size does not matter for cost of capital estimation in some countries. Splitting up the sample unavoidably affects the statistical power of our tests and tends to reduce statistical significance in at least some of the subsamples by default. Leaving aside statistical significance, we find that the average return spread between small and large firms is positive in each of the examined regional subsamples. In addition, we show that size and liquidity distributions also differ across regions. As we contend, such differences may potentially explain why the size effect appears strongly in some regions but less so in others.

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Introduction to the Appendices

In the previous sections, we calculated “risk premia over the risk-free rate” (i.e., excess returns) and “premia over CAPM” (i.e., size premia) using market capitalization portfolios and “Size Factor” portfolios.

In Appendices A through C, we provide detailed exhibits for excess returns and size premia (and various other statistics) for each of the six measures of firm size, plus the seventh measure, the aggregate “Size Factor”. In this updated April 2019 version of the study, we use a fully updated dataset, including return data for the period 1990 – 2019 (of 5,739 firms from 17 European countries).

The seven measures of size used in the exhibits are:

Market Capitalization	Number of common shares outstanding times share price at the beginning of the calendar year.
Market Value of Invested Capital	Market capitalization of equity at the beginning of the calendar year plus the book value of debt of the most recent fiscal year prior to the calendar year.
Book Equity	Ending book value of equity of the most recent fiscal year prior to the calendar year.
Total Assets	Ending book value of total assets of the most recent fiscal year prior to the calendar year.
Sales	Total revenues of the most recent fiscal year prior to the calendar year.
Number of Employees	Number of employees at the end of the most recent fiscal year prior to the calendar year.
Size Factor	A combination of the other six size measures, where: Factor = $-2.8818 +$ $0.0802 \times \ln(\text{Market capitalization}) +$ $0.0842 \times \ln(\text{MV of invested capital}) +$ $0.0869 \times \ln(\text{Book equity}) +$ $0.0877 \times \ln(\text{Total assets}) +$ $0.0868 \times \ln(\text{Sales}) + 0.0845 \times \ln(\text{Employees})$

Appendices A through C contain the following exhibits:

Firm Size Measure	Premia Over the Risk-free Rate (i.e., excess returns)	Premia Over CAPM (i.e., size premia)	Comparative Risk Statistics
Market Capitalization	Exhibit A-1	Exhibit B-1	Exhibit C-1
Book Equity	Exhibit A-2	Exhibit B-2	Exhibit C-2
MV of Invested Capital	Exhibit A-3	Exhibit B-3	Exhibit C-3
Total Assets	Exhibit A-4	Exhibit B-4	Exhibit C-4
Sales	Exhibit A-5	Exhibit B-5	Exhibit C-5
Number of employees	Exhibit A-6	Exhibit B-6	Exhibit C-6
Size Factor	Exhibit A-7	Exhibit B-7	Exhibit C-7

The following table briefly summarizes the methodology used to prepare the exhibits in appendices A through C:

Countries Included	Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, and the United Kingdom
Time Period Analyzed	1990–2018
Data Sources	The intersection of the Datastream and Worldscope databases
Company Types Excluded from Analysis	• Financial service companies • Financially-distressed companies • Companies having illiquid stock • Companies in the early stages of their life cycle
Company Types Included in Analysis	Non start-up, financially healthy companies – the typical company in most investment portfolios (i.e., "Low-Financial-Risk" companies).
Currency	€ Euro
Portfolios	16 portfolios (equal-weighted returns).
Beta Type	Sum Beta

Appendix A. Premia over the Risk-Free Rate

Variables displayed in Exhibits A-1 through A-7 of Appendix A have the following definitions:

Portfolio rank	Firms are ranked from largest to smallest.
Average size	The portfolio average of the size measure (used to construct the portfolios) at the beginning of the latest sample year (i.e., 2018).
Log of size	The natural logarithm of the average size variable.
Number as of 2018	The number of firm-year observations in the portfolio in the latest sample year (i.e., 2018).
Sum beta	The sum of the slopes in a regression of monthly portfolio returns on contemporaneous and lagged monthly MSCI Europe index returns (post-ranking).
Standard deviation of returns	The standard deviation of 29 annual portfolio returns (between 1990 and 2018).
Geometric average return	Geometric average of 29 annual portfolio returns (between 1990 and 2018).
Arithmetic average return	Arithmetic average of 29 annual portfolio returns (between 1990 and 2018).
Arithmetic average risk premium	The difference between the arithmetic average portfolio return and arithmetic average return on annual return on the BofA Merrill Lynch German long-term (5 year +) government bond index.
t-Value of arithmetic average risk premium	t-Value indicating the statistical significance of the difference between a portfolio's arithmetic average risk premium and the arithmetic average risk premium in the top size portfolio.
Smoothed average risk premium	The fitted premium from a 3rd order polynomial regression with the arithmetic average risk premium as the dependent variable and the logarithm of the average classification measure as well as its squared and cubic values as independent variables.

Companies Ranked by Market Capitalization

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Premia Over the Risk-Free Rate (RP_{m+s})

Equity Risk Premium Study: Data through December 31, 2018
Data Smoothing with Regression Analysis
Dependent Variable: Average Premium
Independent Variable: Log of Average Market Capitalization

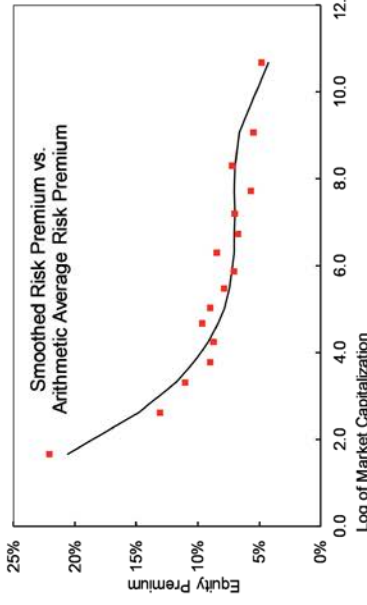
Exhibit A-1

Portfolio Rank by Size	Avg. Mkt Cap (in € millions)	Log of Avg. Mkt Cap	Number as of 2018	Beta (Sum Beta) Since '90	StdDev of Returns	Geometric Avg. Return	Arithmetic Avg. Return	Arithmetic Avg. Risk Premium	t-Value of Arithmetic Risk Premium	Smoothed Avg. Risk Premium
1 (big)	43,019	10.67	130	0.90	17.46%	9.06%	10.95%	4.87%	—	4.25%
2	8,701	9.07	137	1.00	19.85%	9.25%	11.59%	5.53%	0.503	6.62%
3	4,082	8.31	135	1.05	21.48%	10.50%	13.26%	7.20%	1.273	6.93%
4	2,267	7.73	128	1.08	20.53%	9.24%	11.74%	5.69%	0.465	6.99%
5	1,346	7.21	130	1.08	21.60%	10.27%	13.04%	7.00%	1.125	6.98%
6	845	6.74	114	1.03	22.22%	10.04%	12.84%	6.78%	0.873	6.99%
7	547	6.31	128	1.04	21.16%	11.68%	14.51%	8.48%	1.694	7.04%
8	353	5.87	124	1.00	20.65%	10.50%	13.14%	7.09%	1.108	7.19%
9	240	5.48	112	1.00	20.63%	11.28%	13.97%	7.90%	1.346	7.42%
10	154	5.04	107	0.98	20.95%	12.28%	15.08%	9.03%	1.957	7.84%
11	107	4.67	97	0.95	21.77%	12.76%	15.71%	9.68%	2.126	8.35%
12	71	4.25	90	0.97	19.70%	12.19%	14.76%	8.74%	1.904	9.12%
13	43	3.77	81	0.99	21.55%	12.20%	15.03%	8.99%	1.820	10.32%
14	27	3.31	72	1.03	22.63%	13.91%	17.10%	11.08%	2.335	11.82%
15	14	2.62	54	1.03	21.90%	15.81%	19.17%	13.11%	3.392	14.81%
16 (small)	5	1.67	38	1.43	30.33%	22.11%	28.18%	22.11%	3.527	20.62%
Pooled	4,817	8.48	1,677	1.02	20.24%	11.60%	14.22%	8.17%	—	—

Regression Output:

Intercept 36.713%
Log(Size) -12.215%
Log(Size)² 1.670%
Log(Size)³ -0.076%
Adj. R² 92.0%

Smoothed Premium = 36.713% -12.215% * Log(Market Capitalization) + 1.670% * Log(Market Capitalization)² -0.076% * Log(Market Capitalization)³



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Book Equity

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Premia Over the Risk-Free Rate (RP_{m+s})

Equity Risk Premium Study: Data through December 31, 2018
Data Smoothing with Regression Analysis
Dependent Variable: Average Premium
Independent Variable: Log of Average Book Equity

Exhibit A-2

Portfolio Rank by Size	Avg. Book Equity (in € millions)	Log of Avg. Book Equity	Number as of 2018	Beta (Sum Beta) Since '90	StdDev of Returns	Geometric Avg. Return	Arithmetic Avg. Return	Arithmetic Avg. Risk Premium	t-Value of Arithmetic Risk Premium	Smoothed Avg. Risk Premium
1 (big)	19,250	9.87	129	0.95	18.38%	9.68%	11.77%	5.69%	-	5.52%
2	3,178	8.06	132	1.03	19.76%	10.27%	12.68%	6.61%	0.765	6.81%
3	1,537	7.34	132	1.03	20.38%	10.07%	12.58%	6.54%	0.589	6.98%
4	872	6.77	126	1.05	21.23%	9.96%	12.59%	6.53%	0.443	7.06%
5	515	6.24	123	1.06	21.17%	11.56%	14.35%	8.32%	1.642	7.15%
6	339	5.83	120	1.06	21.70%	10.26%	13.07%	7.02%	0.707	7.25%
7	232	5.45	121	1.03	21.65%	10.72%	13.50%	7.43%	0.872	7.39%
8	156	5.05	120	0.99	19.46%	11.73%	14.26%	8.20%	1.449	7.59%
9	109	4.69	106	0.99	20.97%	10.84%	13.54%	7.49%	0.965	7.83%
10	80	4.38	112	0.98	21.04%	11.11%	13.83%	7.81%	1.191	8.11%
11	54	3.98	101	0.96	21.67%	12.01%	14.88%	8.86%	1.689	8.55%
12	35	3.55	98	0.96	20.96%	13.40%	16.22%	10.16%	2.208	9.17%
13	24	3.19	87	0.98	20.87%	12.26%	15.06%	9.03%	2.069	9.81%
14	16	2.76	76	1.01	21.97%	12.77%	15.84%	9.80%	1.768	10.74%
15	9	2.19	60	1.07	21.55%	15.29%	18.54%	12.50%	3.414	12.28%
16 (small)	3	1.18	34	1.37	28.49%	17.56%	22.33%	16.27%	2.666	16.02%
Pooled	2,027	7.61	1,677	1.02	20.24%	11.60%	14.22%	8.17%	-	-

Regression Output:

Intercept 22.331%

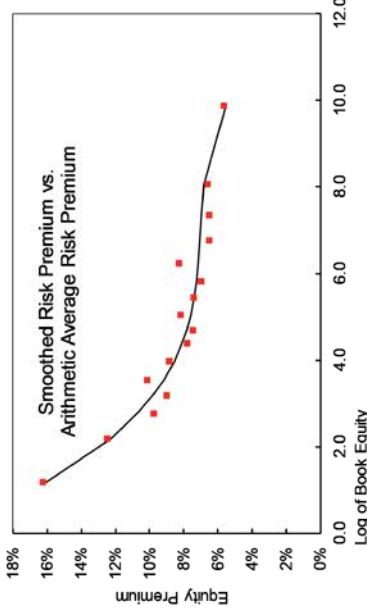
Log(Size) -6.346%

Log(Size)² 0.896%

Log(Size)³ -0.043%

Adj. R² 94.0%

Smoothed Premium = 22.331% - 6.346% * Log(Book Equity) + 0.896% * Log(Book Equity)² - 0.043% * Log(Book Equity)³



Note: "Log" reflects natural logarithm (base e = 2.71828)
Source of underlying data: Refinitiv Reuters Datastream and Worldscope.
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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by MV of Invested Capital

Premia Over the Risk-Free Rate (RP_{m+s})

Exhibit A-3

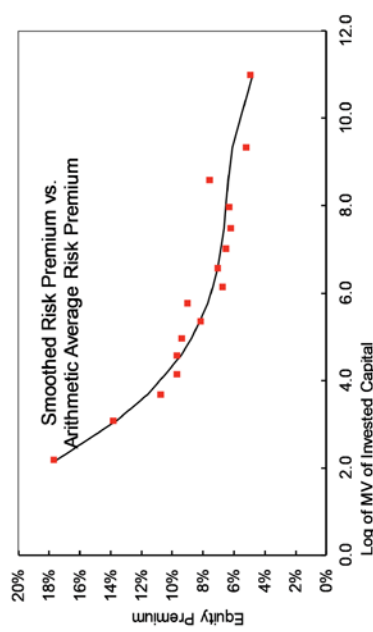
Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018
Equity Risk Premium Study: Data through December 31, 2018
Data Smoothing with Regression Analysis
Dependent Variable: Average Premium
Independent Variable: Log of Average MV of Invested Capital

Portfolio Rank by Size	Avg. MVIC (in € millions)	Log of Avg. MVIC	Number as of 2018	Beta (Sum Beta) Since '90	StdDev of Returns	Geometric Avg. Return	Arithmetic Avg. Return	Arithmetic Avg. Risk Premium	t-Value of Arithmetic Risk Premium	Smoothed Avg. Risk Premium
1 (big)	59,798	11.00	129	0.94	18.14%	9.02%	11.02%	4.93%	-	4.78%
2	11,331	9.34	137	1.01	19.78%	9.00%	11.29%	5.22%	0.216	6.10%
3	5,299	8.58	132	1.07	21.68%	10.84%	13.67%	7.61%	1.554	6.36%
4	2,912	7.98	130	1.08	21.04%	9.78%	12.39%	6.34%	0.792	6.52%
5	1,765	7.48	122	1.05	20.16%	9.78%	12.23%	6.21%	0.700	6.66%
6	1,114	7.02	114	1.06	22.06%	9.77%	12.58%	6.51%	0.775	6.84%
7	712	6.57	123	1.01	20.87%	10.48%	13.08%	7.04%	0.999	7.07%
8	465	6.14	116	1.04	20.81%	10.15%	12.78%	6.72%	0.875	7.37%
9	321	5.77	119	0.95	20.65%	12.25%	15.06%	9.01%	2.050	7.71%
10	210	5.35	104	1.02	21.12%	11.35%	14.24%	8.17%	1.710	8.21%
11	144	4.97	94	0.96	22.06%	12.46%	15.43%	9.40%	1.996	8.76%
12	98	4.58	96	0.95	20.52%	12.99%	15.74%	9.73%	2.707	9.45%
13	63	4.14	83	0.96	20.07%	13.10%	15.74%	9.73%	2.255	10.43%
14	40	3.68	72	1.01	22.18%	13.71%	16.84%	10.79%	2.576	11.64%
15	22	3.08	57	1.00	22.41%	16.35%	19.88%	13.85%	3.905	13.61%
16 (small)	9	2.18	49	1.33	26.95%	19.06%	23.79%	17.73%	3.199	17.51%
Pooled	6,512	8.78	1,677	1.02	20.24%	11.60%	14.22%	8.17%	-	-

Regression Output:

Intercept 32.555%
Log(Size) -9.031%
Log(Size)² 1.070%
Log(Size)³ -0.044%
Adj. R² 95.0%

Smoothed Premium = 32.555% - 9.031% * Log(MV of Invested Capital) + 1.070% * Log(MV of Invested Capital)² - 0.044% * Log(MV of Invested Capital)³



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Exhibit A-4

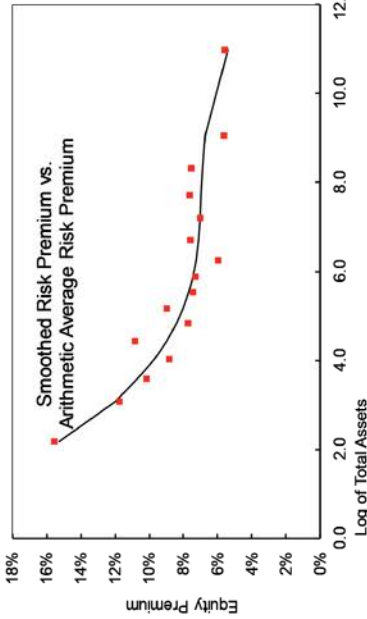
Companies Ranked by Total Assets

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Premia Over the Risk-Free Rate (RP_{m-s})

Equity Risk Premium Study: Data through December 31, 2018
Data Smoothing with Regression Analysis
Dependent Variable: Average Premium
Independent Variable: Log of Average Total Assets

Portfolio Rank by Size	Avg. Total Assets (in € millions)	Log of Avg. Total Assets	Number as of 2018	Beta (Sum Beta) Since '90	StdDev of Returns	Geometric Avg. Return	Arithmetic Avg. Return	Arithmetic Avg. Risk Premium	t-Value of Arithmetic Risk Premium	Smoothed Avg. Risk Premium	Regression Output:
1 (big)	57,887	10.97	127	0.97	18.68%	9.54%	11.68%	5.59%	—	5.40%	Intercept
2	8,615	9.06	129	1.03	19.96%	9.33%	11.68%	5.62%	0.019	6.74%	Log(Size)
3	4,115	8.32	131	1.06	21.57%	10.80%	13.59%	7.53%	1.272	6.88%	Log(Size) ²
4	2,223	7.71	125	1.02	20.04%	11.15%	13.69%	7.64%	1.243	6.95%	Log(Size) ³
5	1,333	7.20	121	1.05	21.05%	10.42%	13.09%	7.04%	0.836	7.01%	Adj. R ²
6	817	6.71	119	1.05	21.11%	10.84%	13.62%	7.59%	1.098	7.12%	Smoothed Premium = 28.082% - 7.854% * Log(Total Assets) + 0.984% * Log(Total Assets) ² - 0.042% * Log(Total Assets) ³
7	517	6.25	113	1.05	21.20%	9.44%	12.03%	5.97%	0.184	7.28%	
8	359	5.88	109	1.02	20.47%	10.76%	13.34%	7.30%	0.888	7.46%	
9	254	5.54	110	1.00	20.58%	10.87%	13.50%	7.44%	1.062	7.70%	
10	176	5.17	114	1.00	21.38%	12.11%	15.01%	8.98%	2.117	8.03%	
11	127	4.84	99	0.92	20.64%	11.18%	13.80%	7.76%	1.163	8.40%	
12	85	4.44	100	0.99	21.72%	13.79%	16.84%	10.83%	2.607	8.97%	
13	56	4.02	86	0.95	20.89%	12.11%	14.86%	8.83%	1.687	9.71%	
14	36	3.58	79	1.01	21.23%	13.24%	16.23%	10.17%	2.441	10.67%	
15	22	3.08	67	1.01	21.99%	14.64%	17.78%	11.73%	2.539	12.00%	
16 (small)	9	2.17	48	1.27	25.78%	17.35%	21.61%	15.56%	3.194	15.25%	
Pooled	5,793	8.66	1,677	1.02	20.24%	11.60%	14.22%	8.17%	—	—	



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Sales

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Premia Over the Risk-Free Rate (RP_{m-s})

Equity Risk Premium Study: Data through December 31, 2018
Data Smoothing with Regression Analysis
Dependent Variable: Average Premium
Independent Variable: Log of Average Sales

Exhibit A-5

Portfolio Rank by Size	Avg. Sales (in € millions)	Log of Avg. Sales	Number as of 2018	Beta (Sum Beta) Since '90	StdDev of Returns	Geometric Avg. Return	Arithmetic Avg. Return	Arithmetic Avg. Risk Premium	t-Value of Arithmetic Risk Premium	Smoothed Avg. Risk Premium
1 (big)	38,003	10.55	125	0.97	18.44%	9.74%	11.86%	5.77%	-	5.56%
2	7,080	8.87	130	1.04	20.12%	10.01%	12.45%	6.40%	0.545	7.18%
3	3,223	8.08	131	1.05	21.16%	10.80%	13.51%	7.46%	1.097	7.41%
4	1,896	7.55	134	1.04	21.18%	11.74%	14.57%	8.52%	1.712	7.47%
5	1,177	7.07	124	1.04	20.00%	10.21%	12.70%	6.64%	0.483	7.49%
6	748	6.62	124	1.04	22.66%	10.92%	13.93%	7.86%	1.109	7.52%
7	476	6.17	118	1.07	21.06%	10.30%	12.96%	6.92%	0.630	7.58%
8	328	5.79	115	1.04	21.25%	11.73%	14.55%	8.51%	1.434	7.66%
9	227	5.42	117	0.99	21.27%	11.72%	14.57%	8.52%	1.511	7.80%
10	161	5.08	109	0.98	19.29%	11.09%	13.50%	7.49%	0.981	7.98%
11	111	4.71	108	0.96	19.89%	10.84%	13.30%	7.26%	0.907	8.25%
12	73	4.29	93	0.95	21.10%	12.59%	15.47%	9.44%	2.253	8.66%
13	47	3.84	76	0.97	21.05%	12.72%	15.56%	9.52%	2.213	9.26%
14	29	3.36	78	0.97	19.63%	12.92%	15.61%	9.56%	1.997	10.10%
15	17	2.81	57	0.98	24.55%	13.80%	17.16%	11.12%	2.007	11.36%
16 (small)	7	1.87	38	1.34	28.07%	16.23%	20.67%	14.60%	2.179	14.34%
Pooled	4,025	8.30	1,677	1.02	20.24%	11.60%	14.22%	8.17%	-	-

Regression Output:

Intercept 24.233%

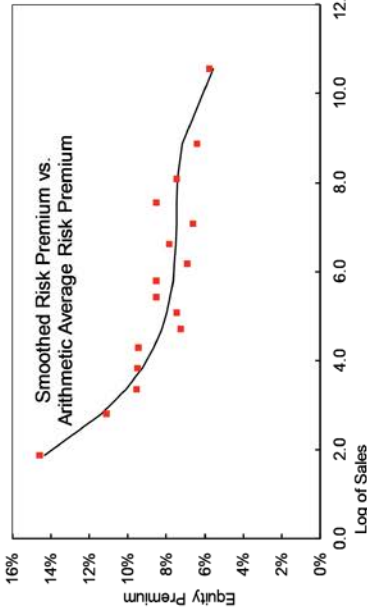
Log(Size) -6.940%

Log(Size)² 0.965%

Log(Size)³ -0.045%

Adj. R² 88.0%

Smoothed Premium = 24.233% - 6.940% * Log(Sales) + 0.965% * Log(Sales)² - 0.045% * Log(Sales)³



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Number of Employees

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

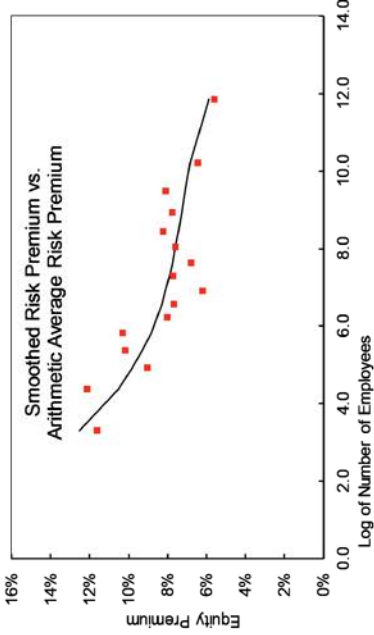
Premia Over the Risk-Free Rate (RP_{m+s})

Equity Risk Premium Study: Data through December 31, 2018
Data Smoothing with Regression Analysis
Dependent Variable: Average Premium
Independent Variable: Log of Average Number of Employees

Exhibit A-6

Portfolio Rank by Size	Avg. # of Employees (in € millions)	Log of Avg. of Employees	Number as of 2018	Beta (Sum Beta) Since '90	StdDev of Returns	Geometric Avg. Return	Arithmetic Avg. Return	Arithmetic Avg. Risk Premium	t-Value of Arithmetic Risk Premium	Smoothed Avg. Risk Premium
1 (big)	140,495	11.85	125	1.03	19.48%	9.44%	11.72%	5.63%	—	5.86%
2	27,139	10.21	124	1.07	21.02%	9.93%	12.55%	6.48%	0.756	6.84%
3	13,203	9.49	132	1.06	21.16%	11.41%	14.18%	8.11%	1.612	7.11%
4	7,490	8.92	128	1.08	21.08%	11.10%	13.83%	7.79%	1.429	7.29%
5	4,638	8.44	129	1.02	20.87%	11.48%	14.29%	8.25%	1.555	7.45%
6	3,089	8.04	130	1.03	22.23%	10.76%	13.70%	7.63%	1.135	7.59%
7	2,069	7.63	119	1.01	19.77%	10.40%	12.84%	6.79%	0.681	7.75%
8	1,469	7.29	113	1.00	20.51%	11.13%	13.76%	7.72%	1.132	7.90%
9	994	6.90	115	0.99	19.54%	9.96%	12.28%	6.23%	0.402	8.10%
10	706	6.56	106	0.98	20.55%	11.12%	13.76%	7.71%	1.310	8.30%
11	507	6.23	94	0.98	19.82%	11.51%	14.04%	8.02%	1.276	8.53%
12	336	5.82	89	0.97	20.73%	13.50%	16.31%	10.32%	2.786	8.86%
13	215	5.37	87	0.96	22.39%	13.10%	16.25%	10.21%	2.570	9.29%
14	137	4.92	69	0.96	20.08%	12.47%	15.10%	9.04%	2.194	9.80%
15	79	4.37	61	1.12	21.15%	14.99%	18.20%	12.15%	2.701	10.55%
16 (small)	27	3.30	56	0.97	26.37%	13.95%	17.68%	11.62%	2.007	12.51%
Pooled	15,111	9.62	1,677	1.02	20.24%	11.60%	14.22%	8.17%	—	—

Smoothed Premium = 23.851% - 4.939% * Log(Number of Employees) + 0.519% * Log(Number of Employees)² - 0.019% * Log(Number of Employees)³



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.
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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Size Factor

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Premia Over the Risk-Free Rate (RP_{m+s})

Equity Risk Premium Study: Data through December 31, 2018

Data Smoothing with Regression Analysis

Dependent Variable: Average Premium

Independent Variable: Average Size Factor

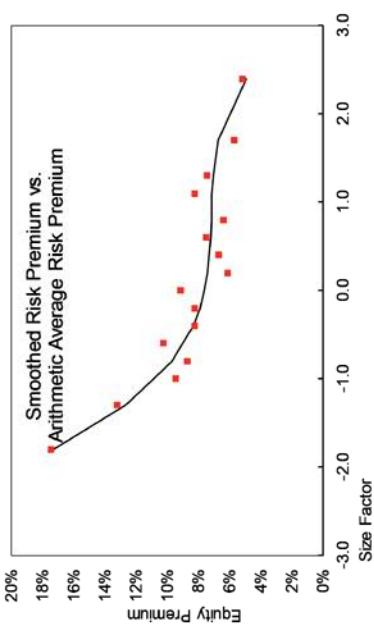
Exhibit A-7

Portfolio Rank by Size	Avg. Size Factor	Log of Avg. Size Factor	Number as of 2018	Beta (Sum Beta) Since '90	StdDev of Returns	Geometric Avg. Return	Arithmetic Avg. Return	Arithmetic Avg. Risk Premium	t-Value of Arithmetic Risk Premium	Smoothed Avg. Risk Premium
1 (big)	2.4	-	127	0.95	18.20%	9.26%	11.30%	5.21%	-	4.93%
2	1.7	-	135	1.05	19.86%	9.42%	11.75%	5.70%	0.445	6.74%
3	1.3	-	131	1.04	20.83%	10.83%	13.51%	7.45%	1.490	7.06%
4	1.1	-	136	1.07	21.29%	11.47%	14.30%	8.24%	1.796	7.14%
5	0.8	-	119	1.06	21.54%	9.75%	12.46%	6.42%	0.657	7.17%
6	0.6	-	123	1.04	21.07%	10.85%	13.58%	7.53%	1.203	7.21%
7	0.4	-	121	1.04	21.27%	10.11%	12.78%	6.72%	0.805	7.29%
8	0.2	-	114	1.00	20.05%	9.76%	12.17%	6.13%	0.503	7.40%
9	0.0	-	116	1.00	22.06%	12.15%	15.23%	9.17%	1.984	7.61%
10	-0.2	-	111	0.97	19.76%	11.69%	14.28%	8.25%	1.815	7.90%
11	-0.4	-	96	1.02	21.22%	11.59%	14.31%	8.28%	1.714	8.32%
12	-0.6	-	98	0.89	19.88%	13.57%	16.30%	10.28%	2.595	9.01%
13	-0.8	-	81	0.95	22.15%	11.84%	14.77%	8.75%	1.938	9.71%
14	-1.0	-	71	1.01	21.77%	12.60%	15.53%	9.47%	1.885	10.88%
15	-1.3	-	63	1.02	21.12%	16.04%	19.30%	13.26%	3.485	12.68%
16 (small)	-1.8	-	35	1.33	28.55%	18.58%	23.57%	17.50%	3.111	17.32%
Pooled	0.4	-	1,677	1.02	20.24%	11.60%	14.22%	8.17%	-	-

Regression Output:

Intercept 7.607%
Size -1.269%
Size² 1.345%
Size³ -0.532%
Adj. R² 89.0%

Smoothed Premium = 7.607% -1.269% * (Size Factor) + 1.345% * (Size Factor)² - 0.532% * (Size Factor)³



Source of underlying data: Refinitiv Reuters Datastream and Worldscope.
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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Appendix B. Premia over CAPM

Variables displayed in Exhibits B-1 through B-7 of Appendix B have the following definitions:

Portfolio rank	Firms are ranked from largest to smallest.
Average size	The portfolio average of the size measure (used to construct the portfolios) at the beginning of the latest sample year (i.e., 2018).
Log of size	The natural logarithm of the average size variable.
Sum beta	The sum of the slopes in a regression of monthly portfolio returns on contemporaneous and lagged monthly MSCI Europe index returns (post-ranking).
Arithmetic average risk premium	The difference between the arithmetic average portfolio return and arithmetic average return on annual return on the BofA Merrill Lynch German long-term (5 year +) government bond index.
t-Value premium over CAPM (sum beta)	t-Value indicating the statistical significance of the difference between a portfolio's premium over CAPM (based on sum beta) and the premium over CAPM in the top size portfolio.
Indicated CAPM premium (sum beta)	The product of the portfolio (sum) beta and the average market risk premium, where the average market risk premium has been set equal to the equity risk premium of market capitalization portfolio 1 (largest firms) divided by the (sum) beta of market capitalization portfolio 1.
Premium over CAPM (sum beta)	The difference between the arithmetic average risk premium and the indicated CAPM premium (based on sum beta).
Smoothed premium over CAPM (sum beta)	The fitted premium from a 3rd order polynomial regression with the premium over CAPM (based on sum beta) as dependent variable and the logarithm of the average classification measure as well as its squared and cubic values as independent variables.

Companies Ranked by Market Capitalization

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Premia over CAPM (Size Premia, RP_s)

Equity Risk Premium Study: Data through December 31, 2018

Data Smoothing with Regression Analysis

Dependent Variable: Premium over CAPM

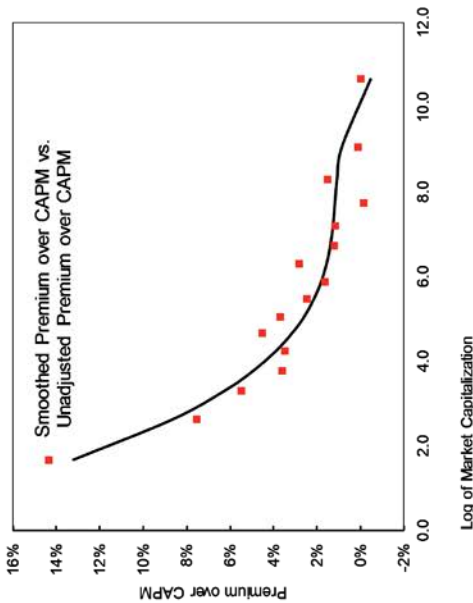
Independent Variable: Log of Average Market Capitalization

Exhibit B-1

Sum Beta

Portfolio Rank by Size	Avg. Mkt Cap (in € millions)	Log of Avg. Mkt Cap	Beta (Sum Beta) Since '90	Arithmetic Avg. Risk Premium	Indicated CAPM Premium	Premium over CAPM	t-Value Premium over CAPM	Smoothed Premium over CAPM
1 (big)	43,018.80	10.67	0.90	4.87%	4.87%	0.00%	-	-0.45%
2	8,700.70	9.07	1.00	5.53%	5.41%	0.12%	0.129	0.87%
3	4,082.00	8.31	1.05	7.20%	5.68%	1.52%	0.881	1.09%
4	2,266.60	7.73	1.08	5.69%	5.84%	-0.15%	-0.029	1.19%
5	1,346.40	7.21	1.08	7.00%	5.84%	1.16%	0.662	1.28%
6	844.90	6.74	1.03	6.78%	5.57%	1.21%	0.588	1.40%
7	547.40	6.31	1.04	8.48%	5.63%	2.85%	1.363	1.57%
8	353.10	5.87	1.00	7.09%	5.41%	1.68%	0.862	1.83%
9	239.90	5.48	1.00	7.90%	5.41%	2.49%	1.124	2.15%
10	154.30	5.04	0.98	9.03%	5.30%	3.73%	1.769	2.62%
11	106.80	4.67	0.95	9.68%	5.14%	4.54%	2.020	3.15%
12	70.50	4.25	0.97	8.74%	5.25%	3.49%	1.730	3.89%
13	43.40	3.77	0.99	8.99%	5.36%	3.63%	1.636	4.96%
14	27.30	3.31	1.03	11.08%	5.57%	5.51%	2.106	6.24%
15	13.70	2.62	1.03	13.11%	5.57%	7.54%	3.136	8.68%
16 (small)	5.30	1.67	1.43	22.11%	7.74%	14.37%	2.988	13.22%

Smoothed Premium = $25.248\% - 8.952\% \cdot \text{Log}(\text{Market Capitalization}) + 1.129\% \cdot \text{Log}(\text{Market Capitalization})^2 - 0.048\% \cdot \text{Log}(\text{Market Capitalization})^3$



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.
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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Book Equity

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Premia over CAPM (Size Premia, RP_s)

Equity Risk Premium Study: Data through December 31, 2018

Data Smoothing with Regression Analysis

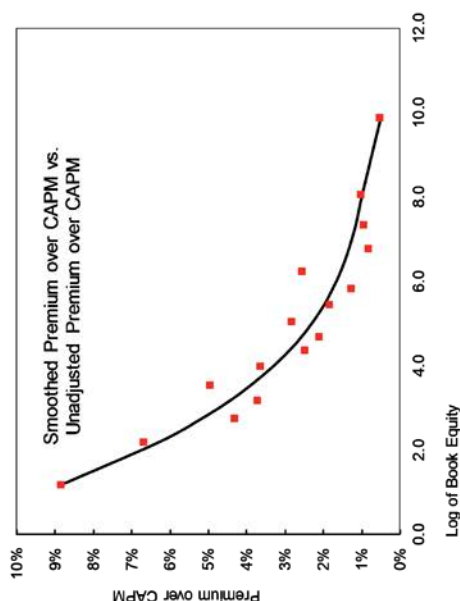
Dependent Variable: Premium over CAPM

Independent Variable: Log of Average Book Equity

Sum Beta

Portfolio Rank by Size	Avg. Book Equity (in € millions)	Log of Avg. Book Equity	Beta (Sum Beta) Since '90	Arithmetic Avg. Risk Premium	Indicated CAPM Premium	Premium over CAPM	t-Value Premium over CAPM	Smoothed Premium over CAPM
1 (big)	19,250.00	9.87	0.95	5.69%	5.14%	0.55%	—	0.51%
2	3,177.60	8.06	1.03	6.61%	5.57%	1.04%	0.379	0.99%
3	1,537.20	7.34	1.03	6.54%	5.57%	0.97%	0.266	1.16%
4	871.70	6.77	1.05	6.53%	5.68%	0.85%	0.132	1.34%
5	514.50	6.24	1.06	8.32%	5.74%	2.58%	1.235	1.55%
6	338.90	5.83	1.06	7.02%	5.74%	1.28%	0.354	1.76%
7	232.00	5.45	1.03	7.43%	5.57%	1.86%	0.641	1.98%
8	155.80	5.05	0.99	8.20%	5.36%	2.84%	1.328	2.27%
9	108.90	4.69	0.99	7.49%	5.36%	2.13%	0.829	2.57%
10	79.70	4.38	0.98	7.81%	5.30%	2.51%	1.083	2.87%
11	53.60	3.98	0.96	8.86%	5.19%	3.67%	1.676	3.32%
12	34.90	3.55	0.96	10.16%	5.19%	4.97%	2.178	3.88%
13	24.30	3.19	0.98	9.03%	5.30%	3.73%	1.954	4.41%
14	15.80	2.76	1.01	9.80%	5.47%	4.33%	1.610	5.15%
15	8.90	2.19	1.07	12.50%	5.79%	6.71%	3.073	6.28%
16 (small)	3.20	1.18	1.37	16.27%	7.41%	8.86%	2.049	8.82%

$$\text{Smoothed Premium} = 12.739\% - 3.807\% \cdot \text{Log}(\text{Book Equity}) + 0.430\% \cdot \text{Log}(\text{Book Equity})^2 - 0.017\% \cdot \text{Log}(\text{Book Equity})^3$$



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.
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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by MV of Invested Capital Premia over CAPM (Size Premia, RP_s) Exhibit B-3

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Equity Risk Premium Study: Data through December 31, 2018

Data Smoothing with Regression Analysis

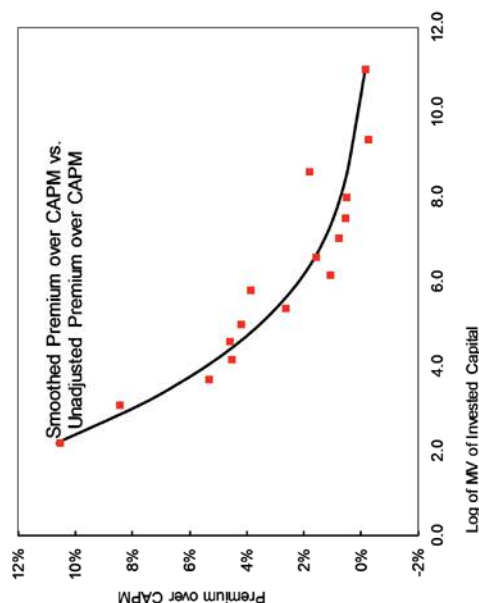
Dependent Variable: Premium over CAPM

Independent Variable: Log of Average MV of Invested Capital

Sum Beta

Portfolio Rank by Size	Avg. MVIC (in € millions)	Log of Avg. MVIC	Beta (Sum Beta) Since '90	Arithmetic Avg. Risk Premium	Indicated CAPM Premium	Premium over CAPM	t-Value Premium over CAPM	Smoothed Premium over CAPM
1 (big)	59,797.90	11.00	0.94	4.93%	5.09%	-0.16%	-	-0.15%
2	11,330.50	9.34	1.01	5.22%	5.47%	-0.25%	-0.089	0.27%
3	5,298.50	8.58	1.07	7.61%	5.79%	1.82%	1.149	0.48%
4	2,912.00	7.98	1.08	6.34%	5.84%	0.50%	0.355	0.72%
5	1,764.80	7.48	1.05	6.21%	5.68%	0.53%	0.367	0.97%
6	1,114.00	7.02	1.06	6.51%	5.74%	0.77%	0.463	1.26%
7	712.20	6.57	1.01	7.04%	5.47%	1.57%	0.827	1.61%
8	465.30	6.14	1.04	6.72%	5.63%	1.09%	0.603	2.02%
9	321.30	5.77	0.95	9.01%	5.14%	3.87%	2.003	2.44%
10	210.10	5.35	1.02	8.17%	5.52%	2.65%	1.465	2.99%
11	144.20	4.97	0.96	9.40%	5.19%	4.21%	1.951	3.56%
12	97.90	4.58	0.95	9.73%	5.14%	4.59%	2.672	4.23%
13	62.60	4.14	0.96	9.73%	5.19%	4.54%	2.198	5.10%
14	39.60	3.68	1.01	10.79%	5.47%	5.32%	2.415	6.15%
15	21.80	3.08	1.00	13.85%	5.41%	8.44%	3.748	7.73%
16 (small)	8.80	2.18	1.33	17.73%	7.20%	10.53%	2.673	10.64%

Smoothed Premium = $20.772\% - 5.776\% \cdot \text{Log}(\text{MV of Invested Capital}) + 0.559\% \cdot \text{Log}(\text{MV of Invested Capital})^2 - 0.019\% \cdot \text{Log}(\text{MV of Invested Capital})^3$



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.
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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Total Assets

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Premia over CAPM (Size Premia, RP_s)

Equity Risk Premium Study: Data through December 31, 2018

Data Smoothing with Regression Analysis

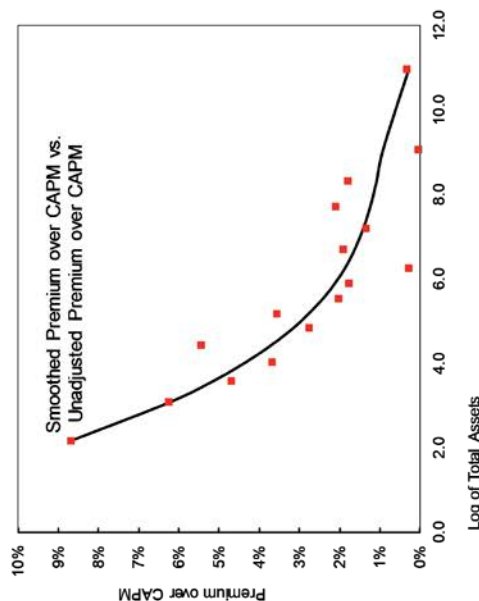
Dependent Variable: Premium over CAPM

Independent Variable: Log of Average Total Assets

Sum Beta

Portfolio Rank by Size	Avg. Total Assets (in € millions)	Log of Avg. Total Assets	Beta (Sum Beta) Since '90	Arithmetic Avg. Risk Premium	Indicated CAPM Premium	Premium over CAPM	t-Value Premium over CAPM	Smoothed Premium over CAPM
1 (big)	57,887.10	10.97	0.97	5.59%	5.25%	0.34%	-	0.27%
2	8,614.70	9.06	1.03	5.62%	5.57%	0.05%	-0.235	0.91%
3	4,115.40	8.32	1.06	7.53%	5.74%	1.79%	0.947	1.07%
4	2,223.00	7.71	1.02	7.64%	5.52%	2.12%	1.074	1.23%
5	1,333.10	7.20	1.05	7.04%	5.68%	1.36%	0.596	1.40%
6	816.50	6.71	1.05	7.59%	5.68%	1.91%	0.852	1.61%
7	516.70	6.25	1.05	5.97%	5.68%	0.29%	-0.016	1.86%
8	358.60	5.88	1.02	7.30%	5.52%	1.78%	0.735	2.12%
9	254.40	5.54	1.00	7.44%	5.41%	2.03%	0.963	2.40%
10	175.50	5.17	1.00	8.98%	5.41%	3.57%	2.033	2.77%
11	127.00	4.84	0.92	7.76%	4.98%	2.78%	1.312	3.14%
12	84.50	4.44	0.99	10.83%	5.36%	5.47%	2.568	3.68%
13	55.70	4.02	0.95	8.83%	5.14%	3.69%	1.744	4.34%
14	35.90	3.58	1.01	10.17%	5.47%	4.70%	2.339	5.14%
15	21.80	3.08	1.01	11.73%	5.47%	6.26%	2.444	6.22%
16 (small)	8.80	2.17	1.27	15.56%	6.87%	8.69%	2.655	8.67%

Smoothed Premium = $17.592\% - 5.255\% \cdot \text{Log}(\text{Total Assets}) + 0.575\% \cdot \text{Log}(\text{Total Assets})^2 - 0.022\% \cdot \text{Log}(\text{Total Assets})^3$



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.
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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Sales

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Premia over CAPM (Size Premia, RP_s)

Equity Risk Premium Study: Data through December 31, 2018

Data Smoothing with Regression Analysis

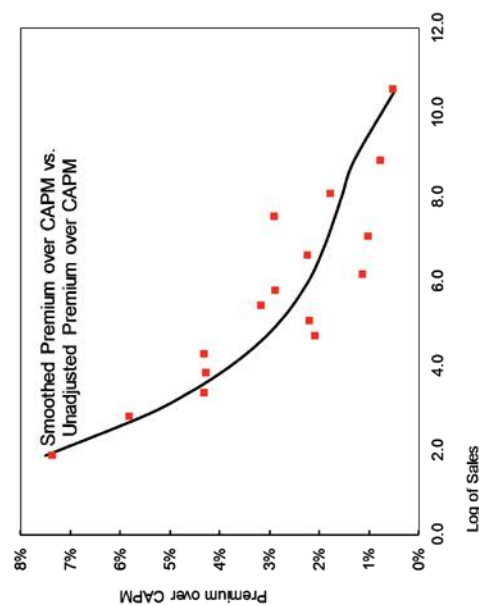
Dependent Variable: Premium over CAPM

Independent Variable: Log of Average Sales

Sum Beta

Portfolio Rank by Size	Avg. Sales (in € millions)	Log of Avg. Sales	Beta (Sum Beta) Since '90	Arithmetic Avg. Risk Premium	Indicated CAPM Premium	Premium over CAPM	t-Value Premium over CAPM	Smoothed Premium over CAPM
1 (big)	38,002.60	10.55	0.97	5.77%	5.25%	0.52%	—	0.47%
2	7,079.80	8.87	1.04	6.40%	5.63%	0.77%	0.171	1.30%
3	3,223.20	8.08	1.05	7.46%	5.68%	1.78%	0.787	1.53%
4	1,896.40	7.55	1.04	8.52%	5.63%	2.89%	1.453	1.68%
5	1,177.00	7.07	1.04	6.64%	5.63%	1.01%	0.238	1.82%
6	747.90	6.62	1.04	7.86%	5.63%	2.23%	0.897	1.97%
7	476.00	6.17	1.07	6.92%	5.79%	1.13%	0.313	2.14%
8	327.60	5.79	1.04	8.51%	5.63%	2.88%	1.214	2.33%
9	226.50	5.42	0.99	8.52%	5.36%	3.16%	1.438	2.54%
10	160.80	5.08	0.98	7.49%	5.30%	2.19%	0.929	2.77%
11	111.40	4.71	0.96	7.26%	5.19%	2.07%	0.929	3.06%
12	73.10	4.29	0.95	9.44%	5.14%	4.30%	2.321	3.46%
13	46.70	3.84	0.97	9.52%	5.25%	4.27%	2.224	3.97%
14	28.80	3.36	0.97	9.56%	5.25%	4.31%	1.983	4.62%
15	16.60	2.81	0.98	11.12%	5.30%	5.82%	1.975	5.52%
16 (small)	6.50	1.87	1.34	14.60%	7.25%	7.35%	1.644	7.50%

Smoothed Premium = 13.421% - 3.981% * Log(Sales) + 0.473% * Log(Sales)² - 0.020% * Log(Sales)³



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Number of Employees

Premia over CAPM (Size Premia, RP_s)

Exhibit B-6

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Equity Risk Premium Study: Data through December 31, 2018

Data Smoothing with Regression Analysis

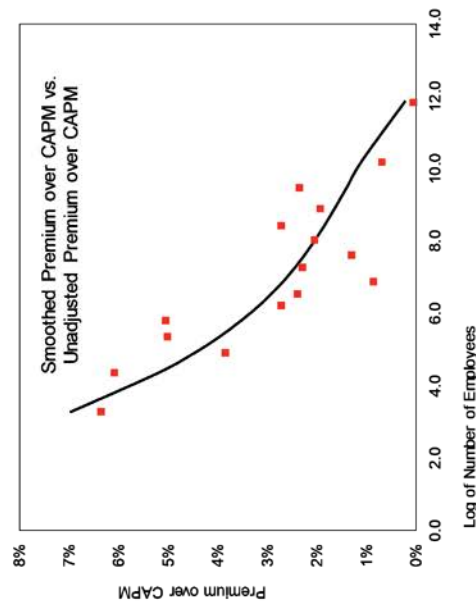
Dependent Variable: Premium over CAPM

Independent Variable: Log of Average Number of Employees

Sum Beta

Portfolio Rank by Size	Avg. # of Employees (in € millions)	Log of Avg. # of Employees	Beta (Sum Beta) Since '90	Arithmetic Avg. Risk Premium	Indicated CAPM Premium	Premium over CAPM	t-Value Premium over CAPM	Smoothed Premium over CAPM
1 (big)	140,494.50	11.85	1.03	5.63%	5.57%	0.06%	—	0.23%
2	27,139.30	10.21	1.07	6.48%	5.79%	0.69%	0.537	1.10%
3	13,203.10	9.49	1.06	8.11%	5.74%	2.37%	1.507	1.40%
4	7,489.50	8.92	1.08	7.79%	5.84%	1.95%	1.241	1.64%
5	4,637.90	8.44	1.02	8.25%	5.52%	2.73%	1.572	1.85%
6	3,088.90	8.04	1.03	7.63%	5.57%	2.06%	1.106	2.04%
7	2,068.80	7.63	1.01	6.79%	5.47%	1.32%	0.726	2.25%
8	1,489.10	7.29	1.00	7.72%	5.41%	2.31%	1.194	2.44%
9	994.40	6.90	0.99	6.23%	5.36%	0.87%	0.516	2.68%
10	705.50	6.56	0.98	7.71%	5.30%	2.41%	1.453	2.92%
11	507.30	6.23	0.98	8.02%	5.30%	2.72%	1.392	3.17%
12	336.20	5.82	0.97	10.32%	5.25%	5.07%	2.953	3.53%
13	215.30	5.37	0.96	10.21%	5.19%	5.02%	2.759	3.97%
14	137.30	4.92	0.96	9.04%	5.19%	3.85%	2.402	4.48%
15	79.40	4.37	1.12	12.15%	6.06%	6.09%	2.497	5.21%
16 (small)	27.10	3.30	0.97	11.62%	5.25%	6.37%	2.100	6.97%

Smoothed Premium = 16.377% - 3.946% * Log(Number of Employees) + 0.377% * Log(Number of Employees)² - 0.013% * Log(Number of Employees)³



Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Size Factor

Historical Equity Risk Premium: Average Since 1990
Data for Year Ending December 31, 2018

Premia over CAPM (Size Premia, RP_s)

Equity Risk Premium Study: Data through December 31, 2018

Data Smoothing with Regression Analysis

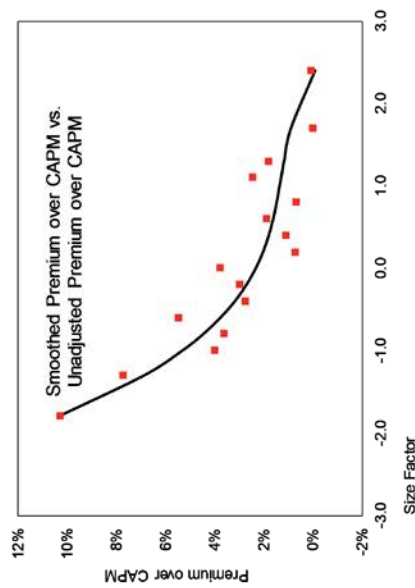
Dependent Variable: Premium over CAPM

Independent Variable: Average Size Factor

Exhibit B-7

Portfolio Rank by Size	Avg. Size Factor	Log of Avg. Size Factor	Sum Beta		Arithmetic Avg. Risk Premium	Indicated CAPM Premium	Premium over CAPM	t-Value Premium over CAPM	Smoothed Premium over CAPM
			Beta (Sum Beta) Since '90						
1 (big)	2.4	—	0.95	5.21%	5.14%	0.07%	—	—	-0.07%
2	1.7	—	1.05	5.70%	5.68%	0.02%	-0.046	—	0.91%
3	1.3	—	1.04	7.45%	5.63%	1.82%	1.150	1.150	1.19%
4	1.1	—	1.07	8.24%	5.79%	2.45%	1.421	1.421	1.31%
5	0.8	—	1.06	6.42%	5.74%	0.68%	0.333	0.333	1.49%
6	0.6	—	1.04	7.53%	5.63%	1.90%	0.934	0.934	1.63%
7	0.4	—	1.04	6.72%	5.63%	1.09%	0.535	0.535	1.81%
8	0.2	—	1.00	6.13%	5.41%	0.72%	0.348	0.348	2.04%
9	0.0	—	1.00	9.17%	5.41%	3.76%	1.833	1.833	2.34%
10	-0.2	—	0.97	8.25%	5.25%	3.00%	1.735	1.735	2.72%
11	-0.4	—	1.02	8.28%	5.52%	2.76%	1.492	1.492	3.19%
12	-0.6	—	0.89	10.28%	4.82%	5.46%	2.752	2.752	3.78%
13	-0.8	—	0.95	8.75%	5.14%	3.61%	1.922	1.922	4.48%
14	-1.0	—	1.01	9.47%	5.47%	4.00%	1.742	1.742	5.33%
15	-1.3	—	1.02	13.26%	5.52%	7.74%	3.314	3.314	6.88%
16 (small)	-1.8	—	1.33	17.50%	7.20%	10.30%	2.600	2.600	10.37%

Smoothed Premium = $2.342\% - 1.680\% * (\text{Size Factor}) + 1.003\% * (\text{Size Factor})^2 - 0.301\% * (\text{Size Factor})^3$



Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

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Appendix C. Comparative Risk Characteristics

Companies Ranked by Market Capitalization				Comparative Risk Characteristics						Exhibit C-1			
Data for Year Ending December 31, 2018				Data Smoothing with Regression Analysis									
				Dependent Variable: Average Premium									
				Independent Variable: Log of Average Market Capitalization									
Portfolio Rank by Size	Portfolio Statistics for 2018			Portfolio Statistics for 1990-2018						StdDev Operating Margin	Avg. of Zero-return Days		
	Avg. Mkt Cap (in € millions)	Log of Avg. Mkt Cap	Number as of 2018	Beta (Sum Beta) Since '90	Unlevered Beta (Sum Beta) Since '90	Arithmetic Avg. Risk Premium	Avg. Debt to MVC	Debt to Value of Equity	Avg. Operating Margin				
1 (big)	43,019	10.67	130	0.90	0.74	4.87%	22.16%	28.47%	14.32%	1.83%	3.97	4.22%	
2	8,701	9.07	137	1.00	0.85	5.53%	18.73%	23.05%	13.03%	2.56%	4.51	4.61%	
3	4,082	8.31	135	1.05	0.90	7.20%	17.90%	21.80%	11.58%	2.19%	4.83	5.42%	
4	2,267	7.73	128	1.08	0.94	5.69%	15.47%	18.30%	11.28%	2.41%	4.75	6.99%	
5	1,346	7.21	130	1.08	0.92	7.00%	17.96%	21.89%	10.29%	2.47%	4.56	6.82%	
6	845	6.74	114	1.03	0.88	6.78%	17.71%	21.52%	9.00%	2.36%	4.80	10.05%	
7	547	6.31	128	1.04	0.89	8.48%	17.61%	21.37%	9.65%	2.87%	4.94	11.81%	
8	353	5.87	124	1.00	0.87	7.09%	15.82%	18.79%	9.59%	3.07%	4.83	13.15%	
9	240	5.48	112	1.00	0.85	7.90%	18.63%	22.90%	8.92%	3.53%	4.88	20.86%	
10	154	5.04	107	0.98	0.86	9.03%	15.82%	18.79%	7.56%	3.28%	4.51	22.25%	
11	107	4.67	97	0.95	0.80	9.68%	19.77%	24.64%	7.66%	4.09%	4.07	25.75%	
12	71	4.25	90	0.97	0.82	8.74%	19.79%	24.67%	6.26%	3.48%	4.20	26.78%	
13	43	3.77	81	0.99	0.83	8.99%	20.67%	26.06%	5.28%	4.25%	4.03	36.29%	
14	27	3.31	72	1.03	0.84	11.08%	22.74%	29.43%	4.82%	6.50%	4.15	36.05%	
15	14	2.62	54	1.03	0.85	13.11%	21.57%	27.50%	4.87%	3.58%	3.12	52.34%	
16 (small)	5	1.67	38	1.43	1.04	22.11%	31.31%	45.58%	3.06%	4.64%	3.12	64.64%	
Note: "Log" reflects natural logarithm (base e = 2.71828)													
Source of underlying data: Refinitiv Reuters Datastream and Worldscope.													
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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)													

Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Book Equity

Data for Year Ending December 31, 2018

Comparative Risk Characteristics

Data Smoothing with Regression Analysis

Dependent Variable: Average Premium

Independent Variable: Log of Average Book Equity

Exhibit C-2

Portfolio Rank by Size	Portfolio Statistics for 2018				Portfolio Statistics for 1990-2018								Average % of Zero-return Days	
	Average Book Equity (in € millions)	Log of Average Book Equity	Number as of 2018	Beta (Sum Beta) Since '90	Unlevered Beta (Sum Beta) Since '90	Arithmetic Average Risk Premium	Average Debt to MVC	Average Debt to Value of Equity	Average Operating Margin	StdDev Operating Margin	Average Z-score	Average % of Zero-return Days		
1 (big)	19,250	9.87	129	0.95	0.74	5.69%	27.84%	38.58%	12.89%	2.05%	3.07	4.91%		
2	3,178	8.06	132	1.03	0.84	6.61%	23.12%	30.07%	12.13%	2.95%	3.87	4.78%		
3	1,537	7.34	132	1.03	0.86	6.54%	20.44%	25.69%	11.12%	2.06%	4.25	6.12%		
4	872	6.77	126	1.05	0.88	6.53%	19.66%	24.47%	11.42%	2.09%	4.28	6.53%		
5	515	6.24	123	1.06	0.91	8.32%	16.92%	20.37%	9.31%	2.40%	4.47	8.80%		
6	339	5.83	120	1.06	0.90	7.02%	18.88%	23.27%	9.99%	2.61%	4.28	11.00%		
7	232	5.45	121	1.03	0.88	7.43%	18.37%	22.50%	9.55%	2.94%	4.88	13.30%		
8	156	5.05	120	0.99	0.85	8.20%	17.12%	20.66%	9.44%	3.09%	5.30	12.28%		
9	109	4.69	106	0.99	0.85	7.49%	17.20%	20.77%	8.29%	2.48%	4.58	18.04%		
10	80	4.38	112	0.98	0.85	7.81%	17.23%	20.82%	8.24%	3.78%	4.77	20.11%		
11	54	3.98	101	0.96	0.84	8.86%	15.19%	17.91%	7.79%	3.04%	5.32	25.86%		
12	35	3.55	98	0.96	0.84	10.16%	15.35%	18.13%	7.30%	4.21%	4.98	32.94%		
13	24	3.19	87	0.98	0.84	9.03%	17.31%	20.93%	7.37%	4.44%	5.12	33.81%		
14	16	2.76	76	1.01	0.88	9.80%	16.32%	19.50%	5.76%	4.96%	5.61	37.32%		
15	9	2.19	60	1.07	0.92	12.50%	17.69%	21.49%	6.43%	5.28%	5.43	40.17%		
16 (small)	3	1.18	34	1.37	1.19	16.27%	15.12%	17.81%	4.11%	4.69%	5.43	54.45%		

Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by MV of Invested Capital

Data for Year Ending December 31, 2018

Comparative Risk Characteristics

Data Smoothing with Regression Analysis

Dependent Variable: Average Premium

Independent Variable: Log of Average MV of Invested Capital

Exhibit C-3

Portfolio Rank by Size	Portfolio Statistics for 2018				Portfolio Statistics for 1990-2018							Average %		
	Average MVIC (in € millions)	Log of Average MVIC	Number as of 2018	Beta (Sum Beta) Since '90	Unlevered Beta (Sum Beta) Since '90	Arithmetic Average Risk Premium	Average Debt to MVIC	Average Debt to Value of Equity	Average Operating Margin	StdDev Operating Margin	Average Z-score	Average of Zero-return Days		
1 (big)	59,798	11.00	129	0.94	0.74	4.93%	26.42%	35.91%	14.39%	1.85%	3.74	4.45%		
2	11,331	9.34	137	1.01	0.84	5.22%	21.09%	26.73%	12.68%	2.81%	4.11	5.01%		
3	5,299	8.58	132	1.07	0.90	7.61%	19.97%	24.95%	11.81%	2.01%	4.48	5.43%		
4	2,912	7.98	130	1.08	0.91	6.34%	18.91%	23.32%	10.58%	2.16%	4.75	6.85%		
5	1,765	7.48	122	1.05	0.90	6.21%	17.93%	21.85%	10.57%	2.62%	4.86	7.28%		
6	1,114	7.02	114	1.06	0.90	6.51%	18.34%	22.46%	9.66%	2.98%	4.82	8.40%		
7	712	6.57	123	1.01	0.88	7.04%	15.99%	19.03%	9.12%	2.32%	4.72	12.56%		
8	465	6.14	116	1.04	0.89	6.72%	17.40%	21.07%	9.43%	2.61%	4.90	12.99%		
9	321	5.77	119	0.95	0.83	9.01%	15.93%	18.95%	8.82%	3.44%	4.85	18.02%		
10	210	5.35	104	1.02	0.87	8.17%	18.18%	22.22%	7.98%	4.27%	5.13	21.76%		
11	144	4.97	94	0.96	0.85	9.40%	14.97%	17.61%	7.85%	3.54%	4.46	24.51%		
12	98	4.58	96	0.95	0.82	9.73%	17.64%	21.42%	6.49%	3.71%	4.16	28.59%		
13	63	4.14	83	0.96	0.79	9.73%	22.23%	28.58%	6.40%	4.04%	4.63	34.89%		
14	40	3.68	72	1.01	0.87	10.79%	17.26%	20.86%	5.10%	5.23%	4.02	37.05%		
15	22	3.08	57	1.00	0.84	13.85%	19.89%	24.83%	4.64%	4.25%	4.20	47.15%		
16 (small)	9	2.18	49	1.33	1.15	17.73%	16.20%	19.33%	3.45%	5.25%	4.20	60.20%		

Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Total Assets

Data for Year Ending December 31, 2018

Comparative Risk Characteristics

Data Smoothing with Regression Analysis

Dependent Variable: Average Premium

Independent Variable: Log of Average Total Assets

Exhibit C-4

Portfolio Rank by Size	Portfolio Statistics for 2018			Portfolio Statistics for 1990-2018								StdDev Operating Margin	Average of Zero-return Days
	Average Total Assets (in € millions)	Log of Average Total Assets	Number as of 2018	Beta (Sum Beta)		Unlevered Beta (Sum Beta)	Arithmetic Average Risk Premium		Average Debt to MVC	Average Debt to Value of Equity	Average Operating Margin		
				Since '90	Since '90		Since '90	Premium					
1 (big)	57,887	10.97	127	0.97	0.72	5.59%	32.41%	47.95%	12.51%	1.85%	3.18	5.03%	
2	8,615	9.06	129	1.03	0.84	5.62%	23.37%	30.50%	11.92%	2.86%	3.14	4.60%	
3	4,115	8.32	131	1.06	0.84	7.53%	25.18%	33.65%	10.44%	2.39%	3.92	6.41%	
4	2,223	7.71	125	1.02	0.86	7.64%	19.47%	24.18%	10.16%	1.92%	4.35	6.56%	
5	1,333	7.20	121	1.05	0.91	7.04%	16.56%	19.85%	10.24%	2.16%	3.97	8.61%	
6	817	6.71	119	1.05	0.88	7.59%	19.47%	24.18%	9.96%	2.59%	4.68	10.62%	
7	517	6.25	113	1.05	0.90	5.97%	17.27%	20.88%	9.72%	2.85%	4.80	11.98%	
8	359	5.88	109	1.02	0.87	7.30%	17.81%	21.67%	9.04%	2.66%	4.66	13.02%	
9	254	5.54	110	1.00	0.87	7.44%	16.40%	19.62%	8.14%	3.17%	4.86	13.64%	
10	176	5.17	114	1.00	0.87	8.98%	16.57%	19.86%	7.89%	2.90%	5.52	22.88%	
11	127	4.84	99	0.92	0.81	7.76%	14.91%	17.52%	8.46%	2.75%	4.83	23.83%	
12	85	4.44	100	0.99	0.85	10.83%	17.21%	20.79%	8.38%	4.76%	5.07	31.40%	
13	56	4.02	86	0.95	0.83	8.83%	16.14%	19.25%	7.40%	4.35%	6.16	33.17%	
14	36	3.58	79	1.01	0.91	10.17%	12.30%	14.03%	7.55%	4.50%	5.92	35.22%	
15	22	3.08	67	1.01	0.90	11.73%	13.58%	15.71%	6.88%	5.38%	7.11	43.19%	
16 (small)	9	2.17	48	1.27	1.20	15.56%	6.73%	7.22%	5.65%	6.59%	7.11	48.25%	

Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Sales

Data for Year Ending December 31, 2018

Comparative Risk Characteristics

Data Smoothing with Regression Analysis
Dependent Variable: Average Premium
Independent Variable: Log of Average Sales

Exhibit C-5

Portfolio Rank by Size	Portfolio Statistics for 2018				Portfolio Statistics for 1990-2018							Average %		
	Average Sales (in € millions)	Log of Average Sales	Number as of 2018	Beta (Sum Beta) Since '90	Unlevered Beta (Sum Beta) Since '90	Arithmetic Premium	Average Debt to MVC	Average Debt to Value of Equity	Average Operating Margin	StdDev Operating Margin	Average Z-score	Average of Zero-return		Days
1 (big)	38,003	10.55	125	0.97	0.74	5.77%	29.70%	42.25%	10.30%	1.65%	3.47	4.91%		
2	7,080	8.87	130	1.04	0.87	6.40%	20.66%	26.04%	9.11%	1.84%	3.40	5.44%		
3	3,223	8.08	131	1.05	0.86	7.46%	22.73%	29.42%	9.70%	1.87%	3.73	6.82%		
4	1,896	7.55	134	1.04	0.87	8.52%	20.67%	26.06%	10.54%	2.59%	4.22	6.39%		
5	1,177	7.07	124	1.04	0.88	6.64%	19.43%	24.12%	11.28%	2.06%	4.60	7.69%		
6	748	6.62	124	1.04	0.90	7.86%	16.77%	20.15%	9.91%	2.45%	4.17	9.15%		
7	476	6.17	118	1.07	0.90	6.92%	19.73%	24.58%	8.25%	2.07%	5.03	13.12%		
8	328	5.79	115	1.04	0.92	8.51%	13.84%	16.06%	9.96%	2.37%	4.88	12.68%		
9	227	5.42	117	0.99	0.85	8.52%	17.72%	21.54%	9.19%	2.80%	4.74	17.84%		
10	161	5.08	109	0.98	0.84	7.49%	17.84%	21.71%	8.23%	3.15%	4.64	19.82%		
11	111	4.71	108	0.96	0.82	7.26%	18.92%	23.33%	9.01%	3.99%	5.01	28.22%		
12	73	4.29	93	0.95	0.84	9.44%	14.76%	17.32%	8.01%	3.03%	5.63	30.54%		
13	47	3.84	76	0.97	0.84	9.52%	16.27%	19.43%	9.64%	4.58%	6.05	33.81%		
14	29	3.36	78	0.97	0.87	9.56%	12.57%	14.38%	8.82%	6.10%	5.54	40.47%		
15	17	2.81	57	0.98	0.84	11.12%	17.95%	21.88%	7.30%	5.95%	6.53	44.56%		
16 (small)	7	1.87	38	1.34	1.24	14.60%	8.67%	9.49%	7.46%	13.48%	6.53	48.85%		

Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Number of Employees

Data for Year Ending December 31, 2018

Comparative Risk Characteristics

Data Smoothing with Regression Analysis
Dependent Variable: Average Premium
Independent Variable: Log of Average Number of Employees

Exhibit C-6

Portfolio Rank by Size	Portfolio Statistics for 2018				Portfolio Statistics for 1990-2018								Average of Zero-return			
	Average # of Employees	Log of Average # of Employees	Number as of 2018	Beta (Sum Beta) Since '90	Unlevered Beta (Sum Beta) Since '90	Arithmetic Average Risk Premium		Average Debt to MVIC	Average Debt to Market Value of Equity	Average Operating Margin	StdDev Operating Margin	Average Z-score	Average % of Zero-return Days			
1 (big)	140,495	11.85	125	1.03	0.80		5.63%	27.94%	38.77%	9.90%	1.40%	3.17	4.82%			
2	27,139	10.21	124	1.07	0.86		6.48%	24.39%	32.26%	10.12%	1.81%	3.63	6.11%			
3	13,203	9.49	132	1.06	0.88		8.11%	21.23%	26.95%	10.20%	2.41%	3.92	6.98%			
4	7,490	8.92	128	1.08	0.92		7.79%	17.96%	21.89%	9.38%	2.11%	4.05	6.99%			
5	4,638	8.44	129	1.02	0.87		8.25%	17.89%	21.79%	10.13%	2.29%	4.41	8.38%			
6	3,089	8.04	130	1.03	0.86		7.63%	20.32%	25.50%	8.55%	1.91%	4.27	10.08%			
7	2,069	7.63	119	1.01	0.88		6.79%	16.64%	19.96%	9.97%	1.98%	4.85	10.00%			
8	1,469	7.29	113	1.00	0.86		7.72%	17.48%	21.18%	8.76%	2.80%	4.67	14.06%			
9	994	6.90	115	0.99	0.86		6.23%	16.25%	19.40%	8.90%	2.58%	4.51	20.25%			
10	706	6.56	106	0.98	0.84		7.71%	17.67%	21.46%	8.67%	3.05%	5.23	18.99%			
11	507	6.23	94	0.98	0.86		8.02%	16.01%	19.06%	8.64%	3.17%	5.10	25.70%			
12	336	5.82	89	0.97	0.84		10.32%	16.26%	19.42%	9.17%	3.48%	5.61	30.99%			
13	215	5.37	87	0.96	0.85		10.21%	14.37%	16.78%	8.76%	4.08%	5.66	35.98%			
14	137	4.92	69	0.96	0.83		9.04%	17.67%	21.46%	9.34%	4.92%	6.21	35.22%			
15	79	4.37	61	1.12	0.98		12.15%	15.14%	17.84%	8.01%	6.07%	6.00	42.58%			
16 (small)	27	3.30	56	0.97	0.84		11.62%	16.93%	20.38%	10.87%	14.21%	6.00	42.90%			

Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

Companies Ranked by Size Factor

Data for Year Ending December 31, 2018

Comparative Risk Characteristics

Data Smoothing with Regression Analysis
Dependent Variable: Average Premium
Independent Variable: Log of Average Size Factor

Exhibit C-7

Portfolio Rank by Size	Portfolio Statistics for 2018			Portfolio Statistics for 1990-2018											
	Average Size Factor	Log of Average Size Factor	Number as of 2018	Beta (Sum Beta)		Unlevered Beta (Sum Beta)		Arithmetic Average Risk Premium		Average Debt to MVIC	Average Debt to Market Value of Equity	Average Operating Margin	StdDev Operating Margin	Average % of Zero-return Days	
				Since '90	Since '90	Since '90	Since '90	Premium	Z-score					Days	
1 (big)	2.4	-	127	0.95	0.73	5.21%	29.54%	41.92%	12.04%	1.74%	3.41	4.89%			
2	1.7	-	135	1.05	0.88	5.70%	20.09%	25.14%	11.39%	2.04%	3.55	4.75%			
3	1.3	-	131	1.04	0.86	7.45%	21.55%	27.47%	11.84%	2.79%	4.08	6.05%			
4	1.1	-	136	1.07	0.89	8.24%	20.19%	25.30%	9.79%	1.88%	4.41	6.10%			
5	0.8	-	119	1.06	0.92	6.42%	16.57%	19.86%	10.31%	2.50%	4.46	7.80%			
6	0.6	-	123	1.04	0.88	7.53%	19.42%	24.10%	9.56%	2.35%	4.92	9.74%			
7	0.4	-	121	1.04	0.91	6.72%	15.48%	18.32%	10.11%	2.48%	5.18	11.04%			
8	0.2	-	114	1.00	0.87	6.13%	16.43%	19.66%	9.98%	2.82%	5.01	11.83%			
9	0.0	-	116	1.00	0.86	9.17%	17.10%	20.63%	8.71%	2.69%	4.25	17.79%			
10	-0.2	-	111	0.97	0.84	8.25%	16.66%	19.99%	6.40%	2.99%	4.99	23.63%			
11	-0.4	-	96	1.02	0.87	8.28%	17.99%	21.94%	8.52%	3.23%	4.98	26.34%			
12	-0.6	-	98	0.89	0.77	10.28%	17.07%	20.58%	8.58%	4.29%	5.78	30.72%			
13	-0.8	-	81	0.95	0.83	8.75%	15.94%	18.96%	8.14%	4.62%	4.62	31.48%			
14	-1.0	-	71	1.01	0.87	9.47%	17.63%	21.40%	6.94%	5.54%	4.93	42.35%			
15	-1.3	-	63	1.02	0.85	13.26%	20.32%	25.50%	5.33%	7.49%	5.56	47.05%			
16 (small)	-1.8	-	35	1.33	1.21	17.50%	10.29%	11.47%	3.66%	6.66%	5.56	56.68%			

Note: "Log" reflects natural logarithm (base e = 2.71828)

Source of underlying data: Refinitiv Reuters Datastream and Worldscope.

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A Study of Return Differences Between Large and Small Companies in Europe (2019 update)

About

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About Rotterdam School of Management, Erasmus University

Rotterdam School of Management, Erasmus University (RSM) is one of Europe's top 10 business schools. RSM provides ground-breaking research and education furthering excellence in all aspects of management and is based in the international port city of Rotterdam – a vital nexus of business, logistics and trade. RSM's primary focus is on developing business leaders with international careers who can become a force for positive change by carrying their innovative mindset into a sustainable future. Our first-class range of bachelor, master, MBA, PhD and executive programmes encourage them to become critical, creative, caring and collaborative thinkers and doers. Study information and activities for future students, executives and alumni are also organised from the RSM office in Chengdu, China. www.rsm.nl

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Updates

This April 2019 version updates the third version of April 2015 (the first version was published in March 2014). The following changes have been made since the first version:

- Update of Part 1
- Update of Exhibits A-7, B-7, and C-7